

AOS-CX 10.08 IP Services Guide

4100i, 6000, 6100 Switch Series



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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 4100i Switch Series (JL817A, JL818A)
- Aruba 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A)
- Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ <code><example-text></code> ■ <code><example-text></code> ■ <i>example-text</i> ■ <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.

Convention	Usage
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment. The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch (CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 4100i Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.

On the 6000 and 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.

ICMP Router Discovery Protocol (IRDP), an extension of the ICMP, is independent of any routing protocol. It allows hosts to discover the IP addresses of neighboring routers that can act as default gateways to reach devices on other IP networks.



On the switches covered by this guide, IRDP is configured on a VLAN interface.

IRDP operation

IRDP uses the following types of ICMP messages:

- Router advertisement (RA): Sent by a router to advertise IP addresses (including the primary and secondary IP addresses) and preference.
- Router solicitation (RS): Sent by a host to request the IP addresses of routers on the subnet.

An interface with IRDP enabled periodically broadcasts or multicasts an RA message to advertise its IP addresses. A receiving host adds the IP addresses to its routing table, and selects the IP address with the highest preference as the default gateway.

When a host attached to the subnet starts up, the host multicasts an RS message to request immediate advertisements. If the host does not receive any advertisements, it retransmits the RS several times. If the host does not discover the IP addresses of neighboring routers because of network problems, the host can still discover them from periodic RAs.

IRDP allows hosts to discover neighboring routers, but it does not suggest the best route to a destination. If a host sends a packet to a router that is not the best next hop, the host will receive an ICMP redirect message from the router.

IP address preference

Every IP address advertised in RAs has a preference value. A larger preference value represents a higher preference. The IP address with the highest preference is selected as the default gateway address.

You can specify the preference for IP addresses to be advertised on a router interface.

An address with the minimum preference value (-2147483648) will not be used as a default gateway address.

Lifetime of an IP address

An RA contains a lifetime field that specifies the lifetime of advertised IP addresses. If the host does not receive a new RA for an IP address within the address lifetime, the host removes the route entry.

All the IP addresses advertised by an interface have the same lifetime.

Advertising interval

A router interface with IRDP enabled sends out RAs randomly between the minimum and maximum advertising intervals. This mechanism prevents the local link from being overloaded by a large number of RAs sent simultaneously from routers.

As a best practice, shorten the advertising interval on a link that suffers high packet loss rates

Destination address of RA

An RA uses either of the following destination IP addresses:

- Broadcast address 255.255.255.255.
- Multicast address 224.0.0.1, which identifies all hosts on the local link.

By default, the destination IP address of an RA is the multicast address. If all listening hosts in a local area network support IP multicast, specify 224.0.0.1 as the destination IP address.

Proxy-advertised IP addresses

By default, an interface advertises its primary and secondary IP addresses. You can specify IP addresses of other gateways for an interface to proxy-advertise.

VRF support

In IP-based computer networks, virtual routing and forwarding (VRF) is a technology that allows multiple instances of a routing table to co-exist within the same router at the same time. Because the routing instances are independent, the same or overlapping IP addresses can be used without conflicting with each other.

IRDP is VRF aware. As the router advertisements and solicit processing occurs on the interface, packet is through the interface and corresponding VRF.

VSX synchronization

IRDP supports VSX synchronization. For more information on using VSX, see the *Virtual Switching Extension (VSX) Guide* for your switch and software version

Configuring IRDP

Prerequisites

A layer 3 interface.

Procedure

1. Enable IRDP on an interface with the command `ip irdp`.
2. Set the maximum hold time with the command `ip irdp holdtime`.
3. Set the maximum router advertisement interval with the command `ip irdp maxadvertinterval`.
4. Set the minimum router advertisement interval with the command `ip irdp minadvertinterval`.
5. Set the IRDP preference level with the command `ip irdp preference`.
6. Review IRDP configuration settings with the command `show ip irdp`.

Example

This example creates the following configuration:

- Enables IRDP on the layer 3 VLAN interface 2 with packet type set to broadcast.
- Sets the hold time to 5000 seconds.
- Sets the advertisement interval to 30 seconds.
- Sets the minimum advertisement interval to 25 seconds.
- Sets the IRDP preference level to 25.

```

switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp broadcast
switch(config-if-vlan)# ip irdp holdtime 5000
switch(config-if-vlan)# ip irdp maxadvertinterval 30
switch(config-if-vlan)# ip irdp minadvertinterval 25
switch(config-if-vlan)# ip irdp preference 25

```

IRDP commands

diag-dump irdp basic

diag-dump irdp basic

Description

Displays diagnostic information for IRDP.

Example

```

switch# diag-dump irdp basic
=====
[Start] Feature irdp Time : Thu Jan  7 04:46:25 2021
=====
-----
[Start] Daemon hpe-rdiscd
-----
Interface: vlan2 (state : Down)
rdisc ipv4 (enabled: 1, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface
Interface: vlan1 (state : Down)
rdisc ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
No advertisable IPv4 addresses on the interface

-----
[End] Daemon hpe-rdiscd
-----
[End] Feature irdp
=====
Diagnostic-dump captured for feature irdp

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

ip irdp

```
ip irdp [broadcast | multicast]
no ip irdp
```

Description

Enables IRDP on an interface and specifies the packet type that is used to send advertisements. By default, the packet type is set to `multicast`. IRDP is only supported on layer 3 interfaces.

The `no` form of this command disables IRDP on an interface.

Parameter	Description
<code>broadcast</code>	Advertisements are sent as broadcast packets to IP address 255.255.255.255.
<code>multicast</code>	Advertisements are sent as multicast packets to the multicast group with IP address 24.0.0.1. Default.

Examples

Enabling IRDP on interface vlan 2 with packet type set to the default value (multicast).

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp
```

Enabling IRDP on interface vlan 2 with packet type set to broadcast.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp broadcast
```

Disabling IRDP.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config-if-vlan</code>	Administrators or local user group members with execution rights for this command.

ip irdp holdtime

```
ip irdp holdtime <TIME>
no ip irdp holdtime <TIME>
```

Description

Specifies the maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives. When a new advertisement arrives, hold time is reset. Hold time must be greater than or equal to the maximum advertisement interval. Therefore, if the hold time for an advertisement expires, the host can reasonably conclude that the router interface that sent the advertisement is no longer available. The default hold time is three times the maximum advertisement interval.

The `no` form of this command removes the specified maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives and update it to the default value.

Parameter	Description
<code><TIME></code>	Specifies the lifetime of router advertisements sent from this interface. Range: 4 to 9000 seconds. Default: 1800 seconds.

Example

Setting the hold time for VLAN interface 2 to 5000 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp holdtime 5000
```

Removing the the hold time for VLAN interface 2 to 5000 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp holdtime 5000
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip irdp maxadvertinterval

```
ip irdp maxadvertinterval <TIME>
no ip irdp maxadvertinterval <TIME>
```

Description

Specifies the maximum router advertisement interval.

The `no` form of this command removes the specified maximum router advertisement interval and reverts to the default value.

Parameter	Description
<i><TIME></i>	Specifies the maximum time allowed between the sending of unsolicited router advertisements. Range: 4 to 1800 seconds. Default: 600 seconds.

Example

Setting the advertisement interval for VLAN interface 2 to 30 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp maxadvertinterval 30
```

Removing the advertisement interval for VLAN interface 2 to 30 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp maxadvertinterval 30
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip irdp minadvertinterval

```
ip irdp minadvertinterval <TIME>
no ip irdp minadvertinterval <TIME>
```

Description

Specifies the minimum amount of time the switch waits between sending router advertisements. By default, this value is automatically set by the switch to be 75% of the value configured for maximum router advertisement interval. Use this command to override the automatically configured value.

The **no** form of this command removes the specified minimum amount of time the switch waits between sending router advertisements and reverts to the default value.

Parameter	Description
<i><TIME></i>	Specifies the minimum time allowed between the sending of unsolicited router advertisements. Range: 3 to 1800 seconds. Default: 450 seconds (75% of the default value for maximum router advertisement interval).

Example

Setting the minimum advertisement interval for VLAN interface 2 to 25 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp minadvertinterval 25
```

Removing the minimum advertisement interval for VLAN interface 2 to 25 seconds:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp minadvertinterval 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ip irdp preference

```
ip irdp preference <LEVEL>
no ip irdp preference <LEVEL>
```

Description

Specifies the IRDP preference level. If a host receives multiple router advertisement messages from different routers, the host selects the router that sent the message with the highest preference as the default gateway.

The **no** form of this command removes the specified IRDP preference level and reverts to the default value.

Parameter	Description
<LEVEL>	Specifies the IRDP preference level. Range: -2147483648 to 2147483647. Default: 0.

Example

Setting the IRDP preference level for VLAN interface 2 to 25.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip irdp preference 25
```

Removing the IRDP preference level for VLAN interface 2 to 25.


```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip irdp preference 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

show ip irdp

show ip irdp

Description

Displays IRDP configuration settings.

Example

```
switch# sh ip irdp

ICMP Router Discovery Protocol

Interface      Status   Advertising Address   Minimum Interval   Maximum Interval   Holdtime   Preference
-----
vlan1          Disabled multicast  450           600           1800           0
bridge_normal  Disabled multicast  450           600           1800           0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

IPv6 RA provides a method for local IPv6 hosts to automatically configure their own IP address (and other settings such as a preferred DNS server) based on information advertised by switches/routers operating on the network.

IPv6 flags

Behavior of IPv6 hosts to IPv6 RA messages is controlled by the managed address configuration flag (M flag), and other stateful configuration flag (O flag).

M flag	O flag	Description
0	0	Indicates that no information is available via DHCPv6.
0	1	Indicates that other configuration information is available via DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.
1	0	Indicates that addresses are available via Dynamic Host Configuration Protocol (DHCPv6).
1	1	If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

Configuring IPv6 RA

Procedure

1. Enable transmission of IPv6 router advertisements with the command `no ipv6 nd suppress-ra`.
2. Optionally, configure IPv6 unicast address prefixes with the command `ipv6 nd prefix`.
3. Optionally, configure support for DNS name resolution with the commands `ipv6 nd ra dns server` and `ipv6 nd ra dns search-list`.
4. For most deployments, the default values for the following features do not need to be changed. If your deployment requires different settings, change the default values with the indicated command:

IPv6 RA setting	Default value	Command to change it
Number of neighbor solicitations to be sent when performing DAD.	1	<code>ipv6 nd dad attempts</code>
Number of neighbor entries in the ND cache.	131072	<code>ipv6 nd cache-limit</code>
Hop limit to be sent in the RA messages.	64	<code>ipv6 nd hop-limit</code>

IPv6 RA setting	Default value	Command to change it
MTU value to be sent in the RA messages.	1500 bytes	<code>ipv6 nd mtu</code>
Neighbor solicitation interval	1000 milliseconds	<code>ipv6 nd ns-interval</code>
Lifetime of a default router.	1800 seconds	<code>ipv6 nd ra lifetime</code>
Retrieval of an IPv6 address by devices.	Disabled	<code>ipv6 nd ra managed-config-flag</code>
Maximum interval between transmissions of IPv6 RAs.	600 seconds	<code>ipv6 nd ra max-interval</code>
Minimum interval between transmissions of IPv6 RAs.	200 seconds	<code>ipv6 nd ra min-interval</code>
Time that an interface considers a device to be reachable.	0 milliseconds (no limit)	<code>ipv6 nd ra reachable-time</code>
Retry period between ND solicitations.	0 (Use locally configured NS-interval)	<code>ipv6 nd ra retrans-timer</code>
Default routing preference for an interface.	Medium	<code>ipv6 nd router-preference</code>

5. Review IPv6 RA configuration settings with the commands `show ipv6 nd interface`, `show ipv6 nd interface prefix`, `show ipv6 nd ra dns server`, and `show ipv6 nd ra dns search-list`.

Example

This example creates the following configuration:

- Enables IPV6 RA on interface `interface vlan 2`.
- Sets the recursive DNS server address to `4001::1` with a lifetime of 400 seconds.
- Sets the minimum interval between transmissions to 3 seconds.
- Sets the maximum interval between transmissions to 13 seconds.
- Sets the lifetime of a default router to 1900 seconds.

```
switch(config)# interface vlan 2
switch(config-if)# no ipv6 nd suppress-ra
switch(config-if)# ipv6 nd ra dns server 4001::1 lifetime 400
switch(config-if)# ipv6 nd ra min-interval 3
switch(config-if)# ipv6 nd ra max-interval 13
switch(config-if)# ipv6 nd ra lifetime 1900
switch(config-if)# end
switch# show ipv6 nd interface vlan 2
Interface vlan2 is up
Admin state is up
IPv6 address:
  2006::1/64 [VALID]
IPv6 link-local address: fe80::98f2:b321:368:6dc6/64 [VALID]
ICMPv6 active timers:
  Last Router-Advertisement sent: 0 Secs
  Next Router-Advertisement sent in: 13 Secs
```

```

Router-Advertisement parameters:
  Periodic interval: 3 to 13 secs
  Router Preference: medium
  Send "Managed Address Configuration" flag: false
  Send "Other Stateful Configuration" flag: false
  Send "Current Hop Limit" field: 64
  Send "MTU" option value: 1500
  Send "Router Lifetime" field: 1900
  Send "Reachable Time" field: 0
  Send "Retrans Timer" field: 0
  Suppress RA: false
  Suppress MTU in RA: true
ICMPv6 error message parameters:
  Send redirects: false
ICMPv6 DAD parameters:
  Current DAD attempt: 1
switch# show ipv6 nd ra dns server
Recursive DNS Server List on: vlan2
  Suppress DNS Server List: No
  DNS Server 1: 2001::1    lifetime 400

```

IPv6 RA scenario

In this scenario, two host computers are auto-configured with IP addresses using IPv6 RA. In addition, the switch provides the hosts with an address of a recursive DNS server.

Procedure

1. Configure the VLAN interfaces with IPv6 addresses.

```

switch# config
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address 2001::1/64
switch(config)# interface vlan 3
switch(config-if-vlan)# ipv6 address 3001::1/64
switch(config)# interface vlan 4
switch(config-if-vlan)# ipv6 address 4001::1/64

```
2. Enable transmission of all IPv6 RA messages.

```

switch(config-if-vlan)# no ipv6 nd suppress-ra

```

IPv6 RA commands

ipv6 address <global-unicast-address>

```

ipv6 address <global-unicast-address>
no ipv6 address <global-unicast-address>

```

Description

Sets a global unicast address on the interface.

The `no` form of this command removes the global unicast address on the interface.



This command automatically creates an IPv6 link-local address on the interface. However, it does not add the `ipv6 address link-local` command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the `ipv6 address link-local` command.

Example

Enabling a global unicast address:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address 3731:54:65fe:2::a7
```

Disabling a global unicast address:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ipv6 address 3731:54:65fe:2::a7
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

ipv6 address autoconfig

```
ipv6 address autoconfig
no ipv6 address autoconfig
```

Description

Enables the interface to automatically obtain an IPv6 address using router advertisement information and the EUI-64 identifier.

The `no` form of this command disables address auto-configuration.

- A maximum of 15 autoconfigured addresses are supported.
- This command automatically creates an IPv6 link-local address on the interface. However, it does not add the `ipv6 address link-local` command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the `ipv6 address link-local` command.

Usage

The IPv6 SLAAC feature lets the router obtain the IPv6 address for the interface it is configured through the SLAAC method. This feature is not available on the `mgmt` VRF.

Example

Enabling unicast autoconfiguring:



```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address autoconfig
```

Disabling unicast autoconfiguring:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ipv6 address autoconfig
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

ipv6 address link-local

ipv6 address link-local [<IPv6-ADDR>/<MASK>]

Description

Enables IPv6 on the current interface. If no address is specified, an IPv6 link-local address is auto-generated for the interface. If an address is specified, auto-configuration is disabled and the specified address/mask is assigned to the interface.

To disable IPv6 link-local on the interface, remove `ipv6 address link-local`, `ipv6 address <global-ipv6-address>`, and `ipv6 address autoconfig` from the interface.



This feature is not available on the management VRF.

Parameter	Description
<IPv6-ADDR>	Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.
<MASK>	Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

Example

Enabling IPv6 link-local on the interface:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address link-local
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

ipv6 nd cache-limit

```
ipv6 nd cache-limit <CACHELIMIT>
no ipv6 nd cache-limit [<CACHELIMIT>]
```

Description

Configures the limit on the number of neighbor entries in the ND cache.

The `no` form of this command sets the cache limit to the default value.

Parameter	Description
<CACHELIMIT>	Specifies the neighbor cache entries limit. Range: 1-131072. Default: 131072.

Examples

Setting the cache limit to 20.

```
switch(config)# ipv6 nd cache-limit 20
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

ipv6 nd dad attempts

```
ipv6 nd dad attempts <NUM-ATTEMPTS>  
no ipv6 nd dad attempts [<NUM-ATTEMPTS>]
```

Description

Configures the number of neighbor solicitations to be sent when performing duplicate address detection (DAD) for a unicast address configured on an interface.

The `no` form of this command sets the number of attempts to the default value.

Parameter	Description
<code>dad attempts <NUM-ATTEMPTS></code>	Specifies the number of neighbor solicitations to send. Range: 0-15. Default: 1.

Examples

```
switch(config)# interface vlan 2  
switch(config-if-vlan)# ipv6 nd dad attempts 5
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd hop-limit

```
ipv6 nd hop-limit <HOPLIMIT>  
no ipv6 nd hop-limit [<HOPLIMIT>]
```

Description

Configures the hop limit to be sent in RAs.

The `no` form of this command resets the hop limit to 0. This reset eliminates the hop limit from the RAs that originate on the interface, so the host determines the hop limit.

Parameter	Description
<code>hop-limit <HOPLIMIT></code>	Specifies the hop limit. Range: 0-255. Default: 64.

Examples


```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd hop-limit 64
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd mtu

```
ipv6 nd mtu <MTU-VALUE>
no ipv6 nd mtu [<MTU-VALUE>]
```

Description

Configures the MTU size to be sent in the RA messages.

The **no** form of this command sets hop limit to the default value.

Parameter	Description
<MTU-VALUE>	Specifies the MTU size. Range: 1280-65535 bytes. Default: 1500 bytes.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd mtu 1300
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ns-interval

```
ipv6 nd ns-interval <TIME>
no ipv6 nd ns-interval [<TIME>]
```

Description

Configures the ND time between DAD neighbor solicitations sent for an unresolved destination, or between duplicate address detection neighbor solicitation requests. Increase this setting when neighbor solicitation retries or failures are occurring, or in a slow (WAN) network.

The `no` form of this command sets the ns-interval to the default value.

Parameter	Description
<TIME>	Specifies the neighbor solicitation interval. Range: 1000-3600000 milliseconds. Default: 1000 milliseconds.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ns-interval 1200
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd prefix

```
ipv6 nd prefix <IPV6-ADDR>/<PREFIX-LEN>
    [no-advertise | [valid <LIFETIME-VALUE> preferred
    <LIFETIME-VALUE>] | no-autoconfig | no-onlink]

no ipv6 nd prefix <IPV6-ADDR>/<PREFIX-LEN> [no-advertise
    | [valid <LIFETIME-VALUE> preferred <LIFETIME-VALUE>
    ] | no-autoconfig | no-onlink]

ipv6 nd prefix default [no-advertise | [valid <LIFETIME-VALUE>
    preferred <LIFETIME-VALUE>] | no-autoconfig | no-onlink]}

no ipv6 nd prefix default [no-advertise | [valid <LIFETIME-VALUE>
    preferred <LIFETIME-VALUE>] | no-autoconfig | no-onlink]}
```

Description

Specifies prefixes for the routing switch to include in RAs transmitted on the interface. IPv6 hosts use the prefixes in RAs to autoconfigure themselves with global unicast addresses. The autoconfigured address of a host is composed of the advertised prefix and the interface identifier in the current link-local address of the host.

By default, advertise, autoconfig, and onlink are set.

The `no` form of this command removes the configuration on the interface.

Parameter	Description
<code><IPV6-ADDR>/<PREFIX-LEN></code>	Specifies the IPv6 prefix to advertise in RA. Format: X:X::X:X/M
<code>default</code>	Specifies apply configuration to all on-link prefixes that are not individually set by the <code>ipv6 ra prefix <IPV6-ADDR>/<PREFIX-LEN></code> command. It applies the same valid and preferred lifetimes, link state, autoconfiguration state, and advertise options to the advertisements sent for all on-link prefixes that are not individually configured with a unique lifetime. This also applies to the prefixes for any global unicast addresses configured later on the same interface. Using <code>default</code> once, and then using it again with any new parameter values results in the new values replacing the former values in advertisements. If <code>default</code> is used without the <code>no-advertise</code>, <code>no-autoconfig</code>, or <code>no-onlink</code> parameter, the advertisement setting for the absent parameter is returned to its default setting.
<code>no-advertise</code>	Specifies do not advertise prefix in RA.
<code>valid <LIFETIME-VALUE></code>	Specifies the total time, in seconds, the prefix remains available before becoming unusable. After preferred-lifetime expiration, any autoconfigured address is deprecated and used only for transactions only before preferred-lifetime expires. If the valid lifetime expires, the address becomes invalid. You can enter a value in seconds or enter <code>valid infinite</code> which sets infinite lifetime. Default: 2,592,000 seconds which is 30 days. Range: 0-4294967294 seconds.
<code>preferred <LIFETIME-VALUE></code>	Specifies the span of time during which the address can be freely used as a source and destination for traffic. This setting must be less than or equal to the corresponding valid-lifetime setting. You can enter a value in seconds or enter <code>preferred infinite</code> which sets infinite lifetime. Default: 604,800 seconds which is seven days. Range: 0-4294967294 seconds.
<code>no-autoconfig</code>	Specifies do not use prefix for autoconfiguration.
<code>no-onlink</code>	Specifies do not use prefix for onlink determination.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd prefix 4001::1/64 valid 30 preferred 10 no-
autoconfig no-onlink
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra dns search-list

```
ipv6 nd ra dns search-list <DOMAIN-NAME> [lifetime <TIME>]
no ipv6 nd ra dns search-list <DOMAIN-NAME>
```

Description

Configures the DNS Search List (DNSSL) to include in Router Advertisements (RAs) transmitted on the interface.

The `no` form of this command removes the DNS Search List from the RAs transmitted on the interface.

Parameter	Description
<DOMAIN-NAME>	Specifies the domain names for DNS queries.
lifetime <TIME>	Specifies lifetime in seconds. Range: 4-4294967295 seconds. Default: 1800 seconds.

Usage

- DNSSL contains the domain names of DNS suffixes or IPv6 hosts to append to short, unqualified domain names for DNS queries.
- Multiple DNS domain names can be added to the DNSSL by using the command repeatedly.
- A maximum of eight server addresses are allowed.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns search-list test.com lifetime 500
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra dns server

```
ipv6 nd ra dns server <IPV6-ADDR> [lifetime <TIME>]
no ipv6 nd ra dns server <IPV6-ADDR>
```

Description

Configures the IPv6 address of a preferred Recursive DNS Server (RDNSS) to be included in Router Advertisements (RAs) transmitted on the interface.

The `no` form of this command removes the configured DNS server from the RAs transmitted on the interface.

Parameter	Description
<code><IPv6-ADDR></code>	Specifies the RDNSS address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.
<code>lifetime <TIME></code>	Specifies IPv6 DNS server lifetime in seconds. Range: 4-4294967295 seconds. Default: 1800 seconds.

Usage

- Including RDNSS information in RAs provides DNS server configuration for connected IPv6 hosts without requiring DHCPv6.
- Multiple servers can be configured on the interface by using the command repeatedly.
- A maximum of eight server addresses are allowed.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns server 2001::1 lifetime 400
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra lifetime

```
ipv6 nd ra lifetime <TIME>
no ipv6 nd ra lifetime [<TIME>]
```

Description

Configures the lifetime, in seconds, for the routing switch to be used as a default router by hosts on the current interface.

The `no` form of this command sets lifetime to the default of 1800 seconds.

Parameter	Description
<code><TIME></code>	Specifies lifetime in seconds of a default router. A setting of 0 for default router lifetime in an RA indicates that the routing switch is not a default router on the interface. Range: 0-9000 seconds. Default: 1800 seconds.

Usage

- A given host on an interface refreshes the default router lifetime for a specific router each time the host receives an RA from that router.
- A specific router ceases to be a default router candidate for a given host if the default router lifetime expires before the host is updated with a new RA from the router.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra lifetime 1200
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config-if-vlan</code>	Administrators or local user group members with execution rights for this command.

ipv6 nd ra managed-config-flag

```
ipv6 nd ra managed-config-flag
no ipv6 nd ra managed-config-flag
```

Description

Controls the M flag setting in RAs the router transmits on the current interface. Enable the M flag to indicate that hosts can obtain IP address through DHCPv6. The M flag is disabled by default.

The `no` form of this command turns off (disables) the M flag.

Usage

- Enabling the M flag directs hosts to acquire their IPv6 addressing for the current interface from a DHCPv6 server.

- When the M-bit is enabled, receiving hosts ignore the O flag setting, which is configured using the command `ipv6 nd ra other-config-flag`.
- When the M-bit is disabled (the default), receiving hosts expect to receive their IPv6 addresses from RA.

M flag	O flag	Description
0	0	Indicates that no information is available via DHCPv6.
0	1	Indicates that other configuration information is available via DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.
1	0	Indicates that addresses are available via Dynamic Host Configuration Protocol (DHCPv6).
1	1	If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra managed-config-flag
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra max-interval

```
ipv6 nd ra max-interval <TIME>
no ipv6 nd ra max-interval [<TIME>]
```

Description

Configures the maximum interval between transmissions of IPv6 RAs on the interface. The interval between RA transmissions on an interface is a random value that changes every time an RA is sent. The interval is calculated to be a value between the current max-interval and min-interval settings.

The `no` form of this command returns the setting to its default, provided the default value is less than the default lifetime value.

Parameter	Description
<code><TIME></code>	Specifies the maximum advertisement time in seconds. Range: 4-1800. Default: 600 seconds.

Usage

- This value has one setting per interface. The setting does not apply to RAs sent in response to a router solicitation received from another device.
- Attempting to set max-interval to a value that is not sufficiently larger than the current min-interval also results in an error message.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra max-interval 30
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra min-interval

```
ipv6 nd ra min-interval <TIME>
no ipv6 nd ra min-interval [<TIME>]
```

Description

Configures the minimum interval between transmissions of IPv6 RAs on the interface. The interval between RA transmissions on an interface is a random value that changes every time an RA is sent. The interval is calculated to be a value between the current max-interval and min-interval settings.

The `no` form of this command returns the setting to its default, provided the default value is less than the current max-interval setting.

Parameter	Description
<code><TIME></code>	Specifies a minimum advertisement time in seconds. Range: 3-1350. Default: 200 seconds.

Usage

- This value has one setting per interface and does not apply to RAs sent in response to a router solicitation received from another device.

- The min-interval must be less than the max-interval. Attempting to set min-interval to a higher value results in an error message.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra min-interval 25
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra other-config-flag

```
ipv6 nd ra other-config-flag
no ipv6 nd ra other-config-flag
```

Description

Controls the O-bit in RAs the router transmits on the current interface; but is ignored unless the M-bit is disabled in RAs. Configure to set the O-bit in RA messages for host to obtain network parameters through DHCPv6. The other-config-flag is disabled by default.

For more information on configuring the M-bit, see `ipv6 nd ra managed-config-flag`.

The `no` form of this command turns off (disables) the setting for this command in RAs.

Usage

Enabling the O-bit while the M-bit is disabled directs hosts on the interface to acquire their other configuration information from DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra other-config-flag
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra reachable-time

```
ipv6 nd ra reachable-time <TIME>
no ipv6 nd ra reachable-time [<TIME>]
```

Description

Sets the amount of time that the interface considers a device to be reachable after receiving a reachability confirmation from the device.

The `no` form of this command sets the reachable time to the default value of 0. (no limit).

Parameter	Description
<TIME>	Specifies the reachable time in milliseconds. Range: 1000-3600000. Default: 0 (no limit).

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra reachable-time 2000
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd ra retrans-timer

```
ipv6 nd ra retrans-timer <TIME>
no ipv6 nd ra retrans-timer [<TIME>]
```

Description

Configures the period (retransmit timer) between ND solicitations sent by a host for an unresolved destination, or between DAD neighbor solicitation requests. By default, hosts on the interface use their own locally configured NS-interval settings instead of using the value received in the RAs.

Increase this timer when neighbor solicitation retries or failures are occur, or in a "slow" (WAN) network.

The `no` form of this command sets the value to the default of 0.

Parameter	Description
<TIME>	Specifies the retransmit timer value in milliseconds. Range: 0 - 4294967295 milliseconds. Default: 0 (Use locally configured NS-interval).

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra retrans-timer 400
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd router-preference

```
ipv6 nd router-preference {high | medium | low}
no ipv6 nd router-preference [high | medium | low]
```

Description

Specifies the value that is set in the Default Router Preference (DRP) field of Router Advertisements (RAs) that the switch sends from an interface. An interface with a DRP value of high will be preferred by other devices on the network over interfaces with an RA value of medium or low.

The `no` form of this command set the value to the default of medium.

Parameter	Description
high	Sets DRP to high.
medium	Sets DRP to medium. Default.
low	Sets DRP to low.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd router-preference high
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

ipv6 nd suppress-ra

```
ipv6 nd suppress-ra [<SUPPRESS-OPTION>]
no ipv6 nd ra suppress-ra [<SUPPRESS-OPTION>]
```

Description

Configures suppression of IPv6 Router Advertisement transmissions on an interface.

The `no` form of this command restores transmission of IPv6 Router Advertisement and options.

Parameter	Description
<code>suppress-ra [<SUPPRESS-OPTION>]</code>	Specifies suppressing RA transmissions. Entering <code>suppress-ra</code> without any options, suppresses all RA messages (default). Or you can enter one of the following options.
<code>dnssl</code>	Specifies suppressing DNSSL options in RA messages.
<code>mtu</code>	Specifies suppressing MTU options in RA messages.
<code>rdnss</code>	Specifies suppressing RDNSS options in RA messages.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd suppress-ra mtu dnssl rdnss
switch(config-if-vlan)# no ipv6 nd suppress-ra mtu dnssl rdnss
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-if-vlan	Administrators or local user group members with execution rights for this command.

show ipv6 nd global traffic

```
show ipv6 nd global traffic
```

Description

Displays IPV6 Neighbor Discovery traffic details on a device.

Examples

```
switch# show ipv6 nd global traffic
ICMPv6 packet Statistics (sent/received)
  Total Messages           :      18/0
  Error Messages           :       0/0
  Destination Unreachables :       0/0
  Time Exceeded            :       0/0
  Parameter Problems       :       0/0
  Echo Request             :       0/0
  Echo Replies             :       0/0
  Redirects                :       0/0
  Packet Too Big           :       0/0
  Router Advertisements    :       4/0
  Router Solicitations     :       0/0
  Neighbor Advertisements  :       0/0
  Neighbor Solicitations   :       3/0
  Duplicate router RA received :    0/0
ICMPv6 MLD Statistics (sent/received)
  V1 Queries :      0/0
  V2 Queries :      0/0
  V1 Reports  :      0/0
  V2 Reports  :     11/0
  V1 Leaves   :      0/0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 nd interface

```
show ipv6 nd interface [<IF-NAME> | all-vrfs | vrf <VRF-NAME>]
```

Description

Displays neighbor discovery information for an interface. If no options are specified, displays information for the default VRF.

Parameter	Description
<IF-NAME>	Displays information about the specified IPv6 enabled interface.
all-vrfs	Displays information about interfaces in all VRFs.
vrf <VRF-NAME>	Displays information about interfaces in a particular VRF. Or, if <VRF-NAME> is not specified, information for the default VRF is displayed.

Examples

Showing information for all VRFs:

```
switch# show ipv6 nd interface all-vrfs

List of IPv6 Interfaces for VRF default
Interface vlan2 is up
  Admin state is up
  IPv6 address:
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent:
    Next Router-Advertisement sent in:
  Router-Advertisement parameters:
    Periodic interval: 200 to 600 secs
    Router Preference: medium
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1800
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
  ICMPv6 error message parameters:
    Send redirects: false
  ICMPv6 DAD parameters:
    Current DAD attempt: 1

List of IPv6 Interfaces for VRF red
Interface vlan3 is up
  Admin state is up
  IPv6 address:
    2001::1/64 [VALID]
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent:
    Next Router-Advertisement sent in:
  Router-Advertisement parameters:
    Periodic interval: 200 to 600 secs
    Router Preference: medium
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1800
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
```

```
    Suppress RA: true
    Suppress MTU in RA: true
ICMPv6 error message parameters:
    Send redirects: false
ICMPv6 DAD parameters:
    Current DAD attempt: 1
```

Showing information for interface vlan 2:

```
switch# show ipv6 nd interface vlan 2
Interface vlan2 is up
  Admin state is up
  IPv6 address:
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent:
    Next Router-Advertisement sent in:
  Router-Advertisement parameters:
    Periodic interval: 200 to 600 secs
    Router Preference: high
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1800
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
  ICMPv6 error message parameters:
    Send redirects: false
  ICMPv6 DAD parameters:
    Current DAD attempt: 1
```

Showing information for the default VRF:

```
switch# show ipv6 nd interface

List of IPv6 Interfaces for VRF default
Interface vlan2 is up
  Admin state is up
  IPv6 address:
    2001::1/64 [VALID]
  IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
  ICMPv6 active timers:
    Last Router-Advertisement sent: 6 Secs
    Next Router-Advertisement sent in: 7 Secs
  Router-Advertisement parameters:
    Periodic interval: 3 to 13 secs
    Router Preference: medium
    Send "Managed Address Configuration" flag: false
    Send "Other Stateful Configuration" flag: false
    Send "Current Hop Limit" field: 64
    Send "MTU" option value: 1500
    Send "Router Lifetime" field: 1900
    Send "Reachable Time" field: 0
    Send "Retrans Timer" field: 0
    Suppress RA: true
    Suppress MTU in RA: true
```

```
ICMPv6 error message parameters:
  Send redirects: false
ICMPv6 DAD parameters:
  Current DAD attempt: 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 nd interface prefix

```
show ipv6 nd interface prefix [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows IPv6 prefix information for all VRFs or a specific VRF. If no options are specified, shows information for the default VRF.

Parameter	Description
all-vrfs	Shows prefix information for all VRFs.
vrf <VRF-NAME>	Name of a VRF.

Examples

Showing prefix information for the default VRF:

```
switch# show ipv6 nd interface prefix

List of IPv6 Interfaces for VRF default
List of IPv6 Prefix advertised on vlan2
  Prefix : 4545::/65
  Enabled : Yes
  Validlife time : 2592000
  Preferred lifetime : 604800
  On-link : Yes
  Autonomous : Yes
```

Showing information for VRF red:

```
switch# show ipv6 nd interface prefix vrf red
```



```
List of IPv6 Interfaces for VRF red
List of IPv6 Prefix advertised on vlan3
Prefix : 2001::/64
Enabled : Yes
Validlife time : 2592000
Preferred lifetime : 604800
On-link : Yes
Autonomous : Yes
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 nd ra dns search-list

```
show ipv6 nd ra dns search-list
```

Description

Displays domain name information on all interfaces.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns search-list test.com
switch# show ipv6 nd ra dns search-list
Recursive DNS Search List on: 1
  Suppress DNS Search List: Yes
  DNS Search 1: test.com    lifetime 1800
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 nd ra dns server

show ipv6 nd ra dns server

Description

Displays DNS server information on all interfaces.

Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns server 2001::1
switch# show ipv6 nd ra dns server
Recursive DNS Server List on: 1
  Suppress DNS Server List: Yes
  DNS Server 1: 2001::1    lifetime 1800
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

sFlow is a technology for monitoring traffic in switched or routed networks. The sFlow monitoring system is comprised of:

- An sFlow Agent that runs on a network device, such as a switch. The agent uses sampling techniques to capture information about the data traffic flowing through the device and forwards this information to an sFlow collector.
- An sFlow Collector that receives monitoring information from sFlow agents. The collector stores this information so that a network administrator can analyze it to understand network data flow patterns. One sFlow collector can receive the data from many sFlow agents.



The sFlow UDP datagrams sent to a collector are not encrypted, therefore any sensitive information contained in an sFlow sample is exposed.

sFlow agent

The sFlow agent on the switch provides ingress sampling of all forwarded layer 2 and layer 3 traffic on LAG and Ethernet ports. High-availability is supported (packet sampling continues to work after switch-over).

The sFlow agent can communicate with up to three sFlow collectors at the same time. The agent communicates with collectors only on the default VRF.

Although you can configure very high sampling rates, the switch may drop samples if it cannot handle the rate of sampled packets. High sampling rates may also cause high CPU usage resulting in control plane performance issues.

A single sFlow datagram sent to a collector contains multiple flow and counter samples. The total number of samples an sFlow datagram can contain varies depending on the settings for header size and maximum datagram size.

Default settings

- sFlow is disabled on all interfaces.
- Collector port: UDP port 6343.
- Sampling rate: 4096.
- Polling interval: 30 seconds.
- Header size: 128 bytes.
- Max datagram size: 1400 bytes.

Supported features

- Global sampling rate
- Interface counters polling
- Agent IP configuration for IPv4 and IPv6
- Header size configuration

- Max datagram size configuration
- Ingress sampling for all forwarded traffic (L2, L3)
- Enable/Disable sFlow per interface
- Support for three remote collectors
- An out-of-band collector can be defined on the management VRF
- A collector can be defined on the non-default VRF
- Sampling on Ethernet and LAG interfaces
- High availability support (sampling continues to work after switch-over)
- Source IP support (setting source IP for sFlow datagrams sent to a remote collector)

Limitations

- No sampling of egress traffic
- Sampling rate cannot be set per interface (global only)
- sFlow is not configurable via SNMP

Configuring the sFlow agent

Procedure

1. Configure one or more sFlow collectors with the command `sflow collector`. This determines where the sFlow agent sends sFlow information.
2. Enable the sFlow agent on all interfaces, or on a specific interface, with the command `sflow`.
3. Define the address of the sFlow agent with the command `sflow agent-ip`.
4. By default, the source IP address for sFlow datagrams is set to the IP address of the outgoing switch interface on which the sFlow client is communicating with a collector. Since the switch can have multiple routing interfaces, datagrams can potentially be sent on different paths at different times, resulting in different source IP addresses for the same client. To resolve this issue, define a single source IP address. For details, see *Single source IP address* in the *Fundamentals Guide*.
5. For most deployments, the default values for the following settings do not need to be changed. If your deployment requires different settings, change the default values with the indicated commands:

sFlow setting	Default value	Command to change it
Rate at which packets are sampled.	1 in every 4096 packets	<code>sflow sampling</code>
Rate at which the switch sends data to an sFlow collector.	30 seconds	<code>sflow polling</code>
Size of the sFlow header.	128 bytes	<code>sflow header-size</code>
Maximum size of an sFlow datagram.	1400 bytes	<code>sflow max-datagram-size</code>

6. Review sFlow configuration settings with the command `show sflow`.

Example

This example creates the following configuration:

- Configures an sFlow collector with the IP address **10.10.20.209** .
- Enables the sFlow agent on all interfaces.
- Defines the sFlow agent IP address to be **10.10.1.5**.

```
switch(config)# sflow collector 10.10.20.209
switch(config)# sflow
switch(config)# sflow agent-ip 10.0.0.1
```

sFlow scenario

In this scenario, two hosts send sFlow traffic through a switch to an sFlow collector.

Procedure

1. Enable sFlow globally.

```
switch# config
switch(config)# sflow
```
2. Set the sFlow agent IP address to **10.10.12.1**.

```
switch(config)# sflow agent-ip 10.10.12.1
```
3. Set the sFlow collector IP address to **10.10.12.2**.

```
switch(config)# sflow collector 18.2.2.2
```
4. Configure sFlow sampling rate and polling interval.

```
switch(config)# sflow sampling 5000
switch(config)# sflow polling 20
```
5. Configure interface **vlan 2** with IP address **10.10.10.1/24**.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.10.1/24
switch(config)# quit
```
6. Configure interface **vlan 3** with IP address **10.10.11.1/24**.

```
switch(config)# interface vlan 3
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.11.1/24
switch(config)# quit
```
7. Configure interface **vlan 4** with IP address **10.10.12.1/24**.

```
switch(config)# interface vlan 4
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.12.1/24
switch(config)# quit
```
8. Verify sFlow configuration

```
switch# show sflow

sFlow Global Configuration
-----

sFlow                               enabled
Collector IP/Port/Vrf               10.10.10.2/6343/default
Agent Address                        10.0.0.1
```

```
Sampling Rate          1024
Polling Interval       30
Header Size            128
Max Datagram Size     1400
```

```
sFlow Status
```

```
-----
```

```
Running - Yes
```

```
sFlow enabled on Interfaces:
```

```
-----
```

```
lag100
```

```
sFlow Statistics
```

```
-----
```

```
Number of Samples      200
```

sFlow scenario 2

In this scenario, two hosts connected to different switches send sFlow traffic to a collector. A LAG is used to connect the two switches.

Procedure

1. Configure switch 1.
 - a. Enable sFlow globally.

```
switch# config
switch(config)# sflow
```
 - b. Set the sFlow agent IP address to **10.10.12.1**.

```
switch(config)# sflow agent-ip 10.10.12.1
```
 - c. Set the sFlow collector IP address to **10.10.12.2**.

```
switch(config)# sflow collector 10.10.12.2
```
 - d. Configure sFlow sampling rate and polling interval.

```
switch(config)# sflow sampling 5000
switch(config)# sflow polling 10
```
 - e. Create VLAN **8**.

```
switch(config)# vlan 8
switch(config-vlan-8)# no shutdown
switch(config)# exit
```
 - f. Define LAG **100** and assign VLAN **vlan 8** to it.

```
switch(config)# interface lag 100
switch(config-lag-if)# no shutdown
switch(config-lag-if)# vlan access 8
```

- ```
switch(config-lag-if)# lacp mode active
```
- g. Configure interface **vlan 5**.
 

```
switch(config)# interface vlan 5
switch(config-if-vlan)# no shutdown
switch(config-lag-if)# no routing
switch(config-if-vlan)# vlan access 8
```
  - h. Configure VLANs **4** and **5** as members of LAG **100**.
 

```
switch# (config)#interface vlan 4
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# lag 100
switch(config-if-vlan)# exit
switch(config)# interface vlan 5
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# lag 100
switch(config-if-vlan)# exit
```
  - i. Configure interface **vlan 5** with IP address **10.10.12.1/24**.
 

```
switch# (config)#interface vlan 5
switch(config-if-vlan)# no shutdown
switch(config-if-vlan)# ip address 10.10.12.1/24
switch(config-if-vlan)# quit
```
  - j. Verify sFlow configuration.

```
switch# show sflow
sFlow Global Configuration

sFlow enabled
Collector IP/Port/Vrf 10.10.10.2/6343/default
Agent Address 10.0.0.1
Sampling Rate 1024
Polling Interval 30
Header Size 128
Max Datagram Size 1400

sFlow Status

Running - Yes

sFlow enabled on Interfaces:

lag100

sFlow Statistics

Number of Samples 200
```

2. Configure switch 2.
  - a. Create VLAN **8**.
 

```
switch(config)# vlan 8
switch(config-vlan-8)# no shutdown
switch(config)# exit
```
  - b. Define LAG **100** and assign VLAN **vlan 8** to it.
 

```
switch(config)# interface lag 100
switch(config-lag-if)# no shutdown
switch(config-lag-if)# vlan access 8
switch(config-lag-if)# lacp mode active
```
  - c. Configure interface **1/1/1**.
 

```
switch(config)# interface 1/1/1
```

- ```
switch(config-if) # no shutdown
switch(config-if) # vlan access 8
```
- d. Configure interface **1/1/2** and **1/1/3** as members of LAG **100**.
- ```
switch# (config)#interface 1/1/2
switch(config-if) # no shutdown
switch(config-if) # lag 100
switch(config-if) # exit
switch(config-if)# interface 1/1/3
switch(config-if) # no shutdown
switch(config-if) # lag 100
switch(config-if) # exit
```

## sFlow agent commands

### clear sflow statistics

```
clear sflow statistics {global | interface <INTERFACE-NAME>}
```

#### Description

This command clears the sFlow sample statistics counter to 0 either globally or for a specific interface.

| Parameter                  | Description                                       |
|----------------------------|---------------------------------------------------|
| global                     | Specifies all interfaces on the switch.           |
| interface <INTERFACE-NAME> | Specifies the name of an interface on the switch. |

#### Examples

Clearing the global sFlow sample statistics counter to 0 globally:

```
switch(config) # clear sflow statistics global
```

Clearing the global sFlow sample statistics counter to 0 for interface **1/1/1**:

```
switch(config) # clear sflow statistics interface 1/1/1
```

#### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

#### Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

### sflow



```
sflow
no sflow
```

## Description

Enables the sFlow agent.

- In the `config` context, this command enables the sFlow agent globally on all interfaces.
- In an `config-if` context, this command enables the sFlow agent on a specific interface. sFlow cannot be enabled on a member of a LAG, only on the LAG.

The sFlow agent is disabled by default.

The `no` form of this command disables the sFlow agent and deletes all sFlow configuration settings, either globally, or for a specific interface.

## Examples

Enabling sFlow globally on all interfaces:

```
switch(config)# sflow
```

Disabling sFlow globally on all interfaces:

```
switch(config)# no sflow
```

Enabling sFlow on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# sflow
```

Disabling sFlow on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no sflow
```

Enabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# sflow
```

Disabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# no sflow
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context     | Authority                                                                          |
|---------------|---------------------|------------------------------------------------------------------------------------|
| All platforms | config<br>config-if | Administrators or local user group members with execution rights for this command. |

## sflow agent-ip

```
sflow agent-ip <IP-ADDR>
no sflow agent-ip [<IP-ADDR>]
```

### Description

Defines the IP address of the sFlow agent to use in sFlow datagrams. This address must be defined for sFlow to function. HPE recommends that the address:

- can uniquely identify the switch
- is reachable by the sFlow collector
- does not change with time

The `no` form of this command deletes the IP address of the sFlow agent. This causes sFlow to stop working and no datagrams will be sent to the sFlow collector.

| Parameter | Description                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IP-ADDR> | Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. The agent address is used to identify the switch in all sFlow datagrams sent to sFlow collectors. It is usually set to an IP address on the switch that is reachable from an sFlow collector. |

### Examples

Setting the agent address to **10.10.10.100**:

```
switch(config)# sflow agent-ip 10.0.0.100
```

Setting the agent address to **2001:0db8:85a3:0000:0000:8a2e:0370:7334**:

```
switch(config)# sflow agent-ip 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Removing the address configuration from the switch, which results in sFlow being disabled:

```
switch(config)# no sflow agent-ip
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## sflow collector

```
sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
no sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
```

## Description

Defines a collector to which the sFlow agent sends data. Up to three collectors can be defined. At least one collector should be defined, and it must be reachable from the switch for sFlow to work.

| Parameter           | Description                                                                                                                                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| collector <IP-ADDR> | Specifies the IP address of a collector in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. |
| port <PORT>         | Specifies the UDP port on which to send information to the sFlow collector. Range: 0 to 65536. Default: 6343.                                                                                                       |
| vrf <VRF>           | Specifies the VRF on which to send information to the sFlow collector. The VRF must be defined on the switch. If no VRF is specified, the default VRF (default) is used.                                            |

## Example

Defining a collector with IP address **10.10.10.100** on UDP port **6400**:

```
switch(config)# sflow collector 10.0.0.1 port 6400
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## sflow disable

sflow disable

### Description

Disables the sFlow agent, but retains any existing sFlow configuration settings. The settings become active if the sFlow agent is re-enabled.

### Example

Disabling sFlow support:

```
switch(config)# sflow disable
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## sflow header-size

```
sflow header-size <SIZE>
no sflow header-size [<SIZE>]
```

### Description

Sets the sFlow header size in bytes.

The `no` form of this command sets the header size to the default value of 128.

| Parameter          | Description                                                               |
|--------------------|---------------------------------------------------------------------------|
| header-size <SIZE> | Specifies the sFlow header size in bytes. Range: 64 to 256. Default: 128. |

### Examples

Setting the header size to **64** bytes:

```
switch(config)# sflow header-size 64
```

Setting the header size to the default value of **128** bytes:

```
switch(config)# no sflow header-size
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## sflow max-datagram-size

```
sflow max-datagram-size <SIZE>
no sflow max-datagram-size [<SIZE>]
```

### Description

Sets the maximum number of bytes that are sent in one sFlow datagram.

The `no` form of this command sets maximum number of bytes to the default value of 1400.

| Parameter                | Description                                                                    |
|--------------------------|--------------------------------------------------------------------------------|
| max-datagram-size <SIZE> | Specifies the maximum datagram size in bytes. Range: 1 to 9000. Default: 1400. |

### Examples

Setting the datagram size to **1000** bytes:

```
switch(config)# sflow max-datagram-size 1000
```

Setting the header size to the default value of **1400** bytes:

```
switch(config)# no sflow max-datagram-size
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## sflow polling

```
sflow polling <INTERVAL>
no sflow polling [<INTERVAL>]
```

### Description

Defines the global polling interval for sFlow in seconds.

The `no` form of this command sets the polling interval to the default value of 30 seconds.

| Parameter  | Description                                                                |
|------------|----------------------------------------------------------------------------|
| <INTERVAL> | Specifies the polling interval in seconds. Range: 10 to 3600. Default: 30. |

### Examples

Setting the polling interval to 10:

```
switch(config)# sflow polling 10
```

Setting the polling interval to the default value.

```
switch(config)# no sflow polling
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## sflow sampling

```
sflow sampling <RATE>
no sflow sampling [<RATE>]
```

### Description

Defines the global sampling rate for sFlow in number of packets. The default sampling rate is 4096, which means that one in every 4096 packets is sampled. A warning message is displayed when the sampling rate is set to less than 4096 and proceeds only after user confirmation.

The `no` form of this command sets the sampling rate to the default value of 4096.

| Parameter       | Description                                                         |
|-----------------|---------------------------------------------------------------------|
| sampling <RATE> | Specifies the sampling rate. Range: 1 to 1000000000. Default: 4096. |

## Examples

Setting the sampling rate to **5000**:

```
switch(config)# sflow sampling 5000
```

Setting the sampling rate to the default:

```
switch(config)# no sflow sampling
```

Setting the sampling rate to **1000**:

```
switch(config)# sflow sampling 1000
Setting the sFlow sampling rate lower than 4096 is not recommended and might
affect system performance.
Do you want to continue [y/n]? y
switch(config)#
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## show sflow

```
show sflow [interface <INTERFACE-NAME>]
```

## Description

Shows sFlow configuration settings and statistics for all interfaces, or for a specific interface

| Parameter                  | Description                                       |
|----------------------------|---------------------------------------------------|
| interface <INTERFACE-NAME> | Specifies the name of an interface on the switch. |

## Examples

Showing sFlow information for all interfaces:

```

switch# show sflow
sFlow Global Configuration

sFlow enabled
Collector IP/Port/Vrf 10.10.10.2/6343/default
Agent Address 10.0.0.1
Sampling Rate 1024
Polling Interval 30
Header Size 128
Max Datagram Size 1400

sFlow Status

Running - Yes

sFlow enabled on Interfaces:

lag100

sFlow Statistics

Number of Samples 200

```

Showing sFlow information for interface **1/1/1**:

```

switch# show sflow interface 1/1/1
sFlow configuration - Interface 1/1/1

sFlow enabled
Sampling Rate 1024
Number of Samples 30
sFlow Sampling Status success

```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | Manager (#)     | Administrators or local user group members with execution rights for this command. |



The Dynamic Host Configuration Protocol (DHCP) enables the automatic assignment of IP addresses and other configuration settings to network devices.

## DHCP client

By default, the switch operates as a DHCP client on VLAN 1 or the management interface allowing it to automatically obtain an IP address from a DHCP server on the network to which it is connected.

## DHCP client commands

### ip dhcp

ip dhcp

### Description

Enables the DHCP client on the management interface enabling the interface to automatically obtain an IP address from a DHCP server on the network. By default, the DHCP client is enabled.

### Examples

This example enables the DHCP client on the management interface.

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip dhcp
switch(config-if-vlan)# no shutdown
```

If the interface is not enabled, you can enable it by entering the `no shutdown` command.

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config-if-vlan  | Administrators or local user group members with execution rights for this command. |

## DHCP relay agent



---

DHCP relay is supported by the Aruba 4100i Switch Series but not the Aruba 6000 or 6100 Switch Series.

---

The function of the DHCP relay agent is to forward the DHCP messages to other subnets so that the DHCP server does not have to be on the same subnet as the DHCP clients. The DHCP relay agent transfers DHCP messages from the DHCP clients located on a subnet without a DHCP server, to other subnets. It also relays answers from DHCP servers to DHCP clients.

### Supported interfaces

The DHCP relay agent is supported on layer 3 interfaces, layer 3 VLAN interfaces, and LAG interfaces. DHCP relay is not supported on the management interface.

### VRF support

The DHCP relay agent is VRF aware and behaves as follows when VRFs are defined on the switch:

- DHCP client requests received on an interface are forwarded to the configured servers via the VRF that the interface is part of.
- DHCP server responses received on an interface are forwarded to the client that is reachable via the VRF that the interface is part of.

### DHCP server interoperation

Both DHCP relay and DHCP server can be configured on the same VRF.

## DHCPv4 relay agent



---

DHCP relay is supported by the Aruba 4100i Switch Series but not the Aruba 6000 or 6100 Switch Series.

---

### Hop count in DHCP requests

When a DHCP client broadcasts request, the DHCP relay agent in the switch receives the packets and forwards them to the DHCP server as unicast requests. During this process, the DHCP relay agent increments the hop count before forwarding DHCP packets to the server. The DHCP server, in turn, includes the hop count in the DHCP header in the response sent back to a DHCP client.

### DHCP relay option 82

Option 82 is called the relay agent information option. When a DHCP relay agent forwards client-originated DHCP packets to a DHCP server, the option 82 field is inserted/replaced, or the packet with this option is dropped. Servers recognizing the relay agent information option may use the information to implement policies for the assignment of IP addresses and other parameters. The relay agent relays the server-to-client replies to the client.

If a second relay agent is configured to add its own option 82 information, it can encapsulate option 82 information in messages from a first relay agent. The DHCP server uses the option 82 information from both relay agents to decide the IP address for the client.

### Inter-VRF DHCP relay

The DHCP relay agent supports anycast gateway using option 82 sub-option 5 (RFC 3527). The DHCP relay discovery packet is filled with the client's gateway IP address in sub-option 5 (discovery packet). The DHCP server uses this information to offer an IP address from the right pool. Pool selection occurs by matching the default gateway configuration settings on the DHCP server with the requested gateway IP address in sub-option 5 in the discovery packet.

The switch uses DHCP relay sub-option 151 to enable DHCP relay to forward discovery and reply packets between VXLAN DHCP clients and DHCP servers even when they are on different overlay or underlay VRFs and the DHCP-server is reachable on the default VRF or one of the overlay VRFs.

In general deployments, a renewal of a DHCP client's IP occurs when the client sends a request to the DHCP server directly. In the case of EVPN VXLAN clients, the DHCP server is not directly reachable. Instead, the renewal request is sent to the DHCP relay. DHCP relay agent fills the option 82 sub-option 11 field in the DHCP discovery packet with the client's gateway IP on the VTEP (which is the relay interface IP address of the VTEP) and the DHCP server returns a DHCP offer reply packet with option 54 set to the DHCP server Identifier. When the reply packet is received by the client, the client uses the IP in option 54 to sent subsequent renewal requests to this IP (VTEP's Relay Interface IP) using sub-option 11 (also known as the Server ID Override Sub-option). Refer to RFC 5107 for more details.

Sub-options 5,11,151,152 are filled in the discover packet, only if a source IP address is defined (using the command `ip source-address`) for the given DHCP server's source VRF. If the server does not understand sub-option 151, then the server will add sub-option 152 in offer packet.

In an inter-VRF situation, when both DHCP relay and DHCP snooping are enabled on the switch with option 82, DHCPv4 clients will not receive an IP address.

## Configuring a BOOTP/DHCP relay gateway

The DHCP relay agent selects the lowest-numbered IP address on the interface to use for DHCP messages. The DHCP server then uses this IP address when it assigns client addresses. However, this IP address may not be the same subnet as the one on which the client needs the DHCP service. This feature provides a way to configure a gateway address for the DHCP relay agent to use for relayed DHCP requests, rather than the DHCP relay agent automatically assigning the lowest-numbered IP address.

## Configuring the DHCPv4 relay agent



---

DHCP relay is supported by the Aruba 4100i Switch Series but not the Aruba 6000 or 6100 Switch Series.

---

### Prerequisites

- An enabled layer 3 interface.

### Procedure

1. The DHCPv4 relay agent is enabled by default. If it was previously disabled, enable it with the command `dhcp-relay`.
2. Configure one or more IP helper addresses with the command `ip helper-address`. This determines where the DHCPv4 agent forwards DHCP requests. IP helper addresses can be configured on layer 3 interfaces, layer 3 VLAN interfaces, and LAG interfaces.
3. If you want to modify the content of forwarded DHCP packets or drop DHCP packets, configure option 82 support with the command `dhcp-relay option 82`.
4. Define the gateway address that the DHCPv4 agent will use with the command [ip bootp-gateway](#).
5. If required, enable the hop count increment feature with the command `dhcp-relay hop-count-increment`.
6. Review DHCPv4 relay agent configuration settings with the commands `show dhcp-relay`, `show ip helper-address`, and `show dhcp-relay bootp-gateway`.

### Example

This example creates the following configuration:

- Enables the DHCPv4 relay agent.
- Enables interface **1/1/1** and assigns an IPv4 address to it. (By default, all interfaces are layer 3 and disabled.)
- Defines an IP helper address of **10.10.20.209** on the interface.
- Enables DHCP option 82 support and replaces all option 82 information with the values from the switch with the switch MAC address as the remote ID.

```
switch(config)# dhcp-relay
switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# ip address 198.51.100.1/24
switch(config-if)# ip helper-address 10.10.20.209
switch(config-if)# exit
switch(config)# dhcp-relay option 82 replace mac
switch# show dhcp-relay
```

DHCP Relay Agent : Enabled  
 DHCP Request Hop Count Increment : Enabled  
 Option 82 : Disabled  
 Response Validation : Disabled  
 Option 82 Handle Policy : replace  
 Remote ID : mac

DHCP Relay Statistics:

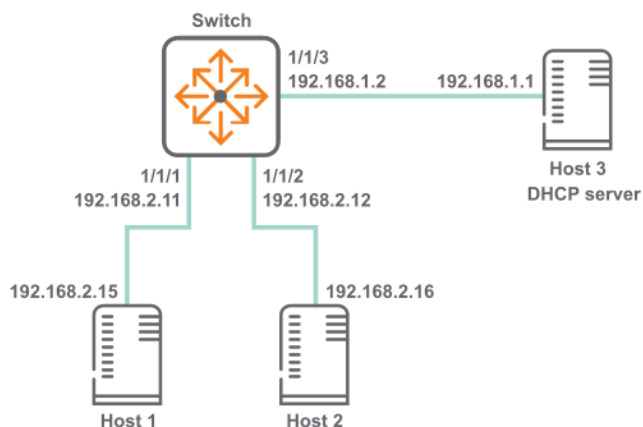
| Valid Requests | Dropped Requests | Valid Responses | Dropped Responses |
|----------------|------------------|-----------------|-------------------|
| 60             | 10               | 60              | 10                |

DHCP Relay Option 82 Statistics:

| Valid Requests | Dropped Requests | Valid Responses | Dropped Responses |
|----------------|------------------|-----------------|-------------------|
| 50             | 8                | 50              | 8                 |

## DHCPv4 relay scenario 1

In this scenario, DHCP relay on the server enables two hosts to obtain their IP addresses from a DHCP server on a different subnet. The physical topology of the network looks like this:



## Procedure

1. DHCP relay is enabled by default. If it was previously disabled, enable it.

```
switch# config
switch(config)# dhcp-relay
```

2. Define an IPv4 helper address on interfaces 1/1/1 and 1/1/2 .

```
switch(config)# interface 1/1/1
switch(config-if)# ip address 192.168.2.11/24
switch(config-if)# ip helper-address 192.168.1.1
switch(config-if)# interface 1/1/2
switch(config-if)# ip address 192.168.2.12/24
switch(config-if)# ip helper-address 192.168.1.1
switch(config-if)# quit
```

3. Verify DHCP relay configuration.

```
switch# show dhcp-relay
```

```
DHCP Relay Agent : Enabled
DHCP Request Hop Count Increment : Enabled
L2VPN Clients : Disabled
Option 82 : Disabled
Source-Interface : Disabled
Response Validation : Disabled
Option 82 Handle Policy : replace
Remote ID : mac
```

DHCP Relay Statistics:

| Valid Requests | Dropped Requests | Valid Responses | Dropped Responses |
|----------------|------------------|-----------------|-------------------|
| 60             | 10               | 60              | 10                |

DHCP Relay Option 82 Statistics:

| Valid Requests | Dropped Requests | Valid Responses | Dropped Responses |
|----------------|------------------|-----------------|-------------------|
| 50             | 8                | 50              | 8                 |

```
switch# show ip helper-address
```

IP Helper Addresses

Interface: 1/1/1

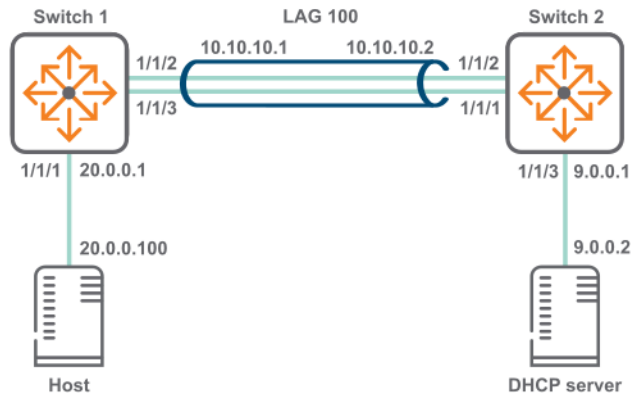
| IP Helper Address | VRF     |
|-------------------|---------|
| 192.168.1.1       | default |

Interface: 1/1/2

| IP Helper Address | VRF     |
|-------------------|---------|
| 192.168.1.1       | default |

## DHCPv4 relay scenario 2

In this scenario, host on switch 1 reaches the DHCP server on switch two via a LAG. The physical topology of the network looks like this:



## Procedure

1. On switch 1:
  - a. Create LAG **100** and assign an IP address to it.
 

```
switch# config
switch(config)# interface lag 100
switch(config-lag-if)# ip address 10.0.10.1/24
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# exit
switch(config)#
```
  - b. Assign an IP address to interface **1/1/1** and a an IP helper address to reach the DHCP server.
 

```
switch(config)# interface 1/1/1
switch(config-if)# ip address 20.0.0.1/8
switch(config-if)# ip helper-address 9.0.0.2
```
  - c. Assign interfaces **1/1/2** and **1/1/3** to LAG **100**

```
switch(config-if)# interface 1/1/2
switch(config-if)# lag 100
switch(config-if)# interface 1/1/3
switch(config-if)# lag 100
switch(config-if)# exit
switch(config)#
```
  - d. Create a route between **10.0.10.2** and **9.0.0.0**.
 

```
switch(config)# ip route 9.0.0.0/24 10.0.10.2
```
2. On switch 2:
  - a. Create LAG **100** and assign an IP address to it.
 

```
switch# config
switch(config)# interface lag 100
switch(config-lag-if)# ip address 10.0.10.2/24
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# exit
switch(config)#
```
  - b. Assign interfaces **1/1/1** and **1/1/2** to LAG **100**

```
switch(config-if)# interface 1/1/2
switch(config-if)# lag 100
switch(config-if)# interface 1/1/3
switch(config-if)# lag 100
switch(config-if)# exit
switch(config)#
```
  - c. Assign an IP address to interface **1/1/3**.
 

```
switch(config)# interface 1/1/3
switch(config-if)# ip address 9.0.0.1/24
```
  - d. Create a route between **20.0.0.0** and **10.0.10.1**.
 

```
switch(config)# ip route 20.0.0.0/8 10.0.10.1
```

## DHCPv4 relay commands

## dhcp-relay

```
dhcp-relay
no dhcp-relay
```

### Description

Enables DHCP relay support. DHCP relay is enabled by default. DHCP relay is not supported on the management interface.

The `no` form of this command disables DHCP relay support.

### Examples

This example enables DHCP relay support.

```
switch(config)# dhcp-relay
```

This example removes DHCP relay support.

```
switch(config)# no dhcp-relay
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcp-relay hop-count-increment

```
dhcp-relay hop-count-increment
no dhcp-relay hop-count-increment
```

### Description

Enables the DHCP relay hop count increment feature, which causes the DHCP relay agent to increment the hop count in all relayed DHCP packets. Hop count is enabled by default.

The `no` form of this command disables the hop count increment feature.

### Examples

Enabling the hop count increment feature.

```
switch(config)# dhcp-relay hop-count-increment
```

Disabling the hop count increment feature.

```
switch(config)# no dhcp-relay hop-count-increment
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

### dhcp-relay l2vpn-clients

```
dhcp-relay l2vpn-clients
no dhcp-relay l2vpn-clients
```

## Description

Enables forwarding of packets from L2 VPN clients. Forwarding is enabled by default.

The `no` form of this command disables forwarding of packets from L2 VPN clients.

## Example

Enabling forwarding of packets from L2 VPN clients.

```
switch(config)# dhcp-relay l2vpn-clients
switch(config)# no dhcp-relay l2vpn-clients
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

### dhcp-relay option 82

```
dhcp-relay option 82 {replace [validate] | drop [validate] |
 keep | source-interface | validate [replace | drop]} [ip | mac]
no dhcp-relay option 82 {replace [validate] | drop [validate] |
 keep | source-interface | validate [replace | drop]} [ip | mac]
```

## Description



Configures the behavior of DHCP relay option 82. A DHCP relay agent can receive a message from another DHCP relay agent having option 82. The relay information from the previous relay agent is replaced by default.

The `no` form of this command disables the DHCP relay option 82 configurations.

| Parameter                     | Description                                                                                                                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>replace</code>          | Replace the existing option 82 field in an inbound client DHCP packet with the information from the switch. The remote ID and circuit ID information from the first relay agent is lost. Default. |
| <code>validate</code>         | Validate option 82 information in DHCP server responses and drop invalid responses.                                                                                                               |
| <code>drop</code>             | Drop any inbound client DHCP packet that contains option 82 information.                                                                                                                          |
| <code>keep</code>             | Keep the existing option 82 field in an inbound client DHCP packet. The remote ID and circuit ID information from the first relay agent is preserved.                                             |
| <code>source-interface</code> | Configures the DHCP relay to use a configured source IP address for inter-VRF server reachability. Set the source IP address with the command <code>ip source-interface</code> .                  |
| <code>ip</code>               | Use the IP address of the interface on which the client DHCP packet entered the switch as the option 82 remote ID.                                                                                |
| <code>mac</code>              | Use the MAC address of the switch as the option 82 remote ID. Default.                                                                                                                            |

## Example

This example enables DHCP option 82 support and replaces all option 82 information with the values from the switch, with the switch MAC address as the remote ID.

```
switch(config)# dhcp-relay option 82 replace mac
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context     | Authority                                                                          |
|-----------|---------------------|------------------------------------------------------------------------------------|
| 4100i     | <code>config</code> | Administrators or local user group members with execution rights for this command. |

## ip bootp-gateway

```
ip bootp-gateway <IPV4-ADDR>
no ip bootp-gateway <IPV4-ADDR>
```

## Description

Configures a gateway address for the DHCP relay agent to use for DHCP requests. By default DHCP relay agent picks the lowest-numbered IP address on the interface.

The `no` form of this command removes the gateway address.

| Parameter                      | Description                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>&lt;IPv4-ADDR&gt;</code> | Specifies the IP address of the gateway in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. |

## Examples

Sets the IP address of the gateway for interface **1/1/1** to **10.10.10.10**.

```
switch(config)# interface 1/1/1
switch(config-if)# ip bootp-gateway 10.10.10.10
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config-if       | Administrators or local user group members with execution rights for this command. |

## ip helper-address

```
ip helper-address <IPv4-ADDR> [vrf <VRF-NAME>]
no ip helper-address <IPv4-ADDR> [vrf <VRF-NAME>]
```

## Description

Defines the address of a remote DHCP server or DHCP relay agent. Up to eight addresses can be defined. The DHCP agent forwards DHCP client requests to all defined servers.

This command requires that you define a source IP address for DHCP relay with the command `ip source-interface`. The configured source IP on the VRF is used to forward DHCP packets to the server.

A helper address cannot be defined on the OOBM interface.

The `no` form of this command removes an IP helper address.

| Parameter                                     | Description                                                                                          |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------|
| <code>helper-address &lt;IPv4-ADDR&gt;</code> | Specifies the helper IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. |
| <code>vrf &lt;VRF-NAME&gt;</code>             | Specifies the name of a VRF. Default: default.                                                       |

## Examples

Defining the IP helper address **10.10.10.209** on interface **1/1/1**.

```
switch(config)# interface 1/1/1
switch(config-if)# ip helper-address 10.10.10.209
```

Removing the IP helper address **10.10.10.209** on interface **1/1/1**.

```
switch(config-if)# no ip helper-address 10.10.10.209
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config-if       | Administrators or local user group members with execution rights for this command. |

## show dhcp-relay

show dhcp-relay

## Description

Shows DHCP relay configuration settings.

## Example

```
switch# show dhcp-relay

DHCP Relay Agent : Enabled
DHCP Request Hop Count Increment : Enabled
L2VPN Clients : Disabled
Option 82 : Disabled
Source-Interface : Disabled
Response Validation : Disabled
Option 82 Handle Policy : replace
Remote ID : mac

DHCP Relay Statistics:

Valid Requests Dropped Requests Valid Responses Dropped Responses

60 10 60 10

DHCP Relay Option 82 Statistics:

Valid Requests Dropped Requests Valid Responses Dropped Responses

50 8 50 8
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                                                                                                              |
|-----------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Manager (#)     | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show dhcp-relay bootp-gateway

show dhcp-relay bootp-gateway [interface <INTERFACE-NAME>]

## Description

Shows the bootp gateway defined for all interfaces or a specific interface.

| Parameter        | Description                                       |
|------------------|---------------------------------------------------|
| <INTERFACE-NAME> | Specifies an interface. Format: member/slot/port. |

## Examples

```
switch# show dhcp-relay bootp-gateway
```

```
BOOTP Gateway Entries
```

```
Interface Source IP

1/1/1 1.1.1.1
1/1/2 1.1.1.2
```

```
switch# show ip helper-address interface 1/1/1
```

```
BOOTP Gateway Entries
```

```
Interface Source IP

1/1/1 1.1.1.1
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                                                                                                              |
|-----------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Manager (#)     | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ip helper-address

show ip helper-address [interface <INTERFACE-ID>]

## Description

Shows the helper IP addresses defined for all interfaces or a specific interface.

| Parameter                | Description                                       |
|--------------------------|---------------------------------------------------|
| interface <INTERFACE-ID> | Specifies an interface. Format: member/slot/port. |

## Example

```
switch#
IP Helper Addresses

Interface: 1/1/1
IP Helper Address VRF

192.168.20.1 default
192.168.10.1 default

Interface: 1/1/2
IP Helper Address VRF

192.168.30.1 RED
```

```
switch# show ip helper-address interface 1/1/1
IP Helper Addresses

Interface: 1/1/1
IP Helper Address VRF

192.168.20.1 default
192.168.10.1 default
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                    |
|-----------|-----------------|--------------------------------------------------------------|
| 4100i     | Manager (#)     | Operators or Administrators or local user group members with |

| Platforms | Command context | Authority                                                                                                 |
|-----------|-----------------|-----------------------------------------------------------------------------------------------------------|
|           |                 | execution rights for this command. Operators can execute this command from the operator context (>) only. |

## DHCPv6 relay agent



DHCP relay is supported by the Aruba 4100i Switch Series but not the Aruba 6000 or 6100 Switch Series.

### Supporting VXLAN topologies or inter-VRF deployment

When deploying EVPN VXLAN or inter-VRF topologies where the source VRFs for the DHCP and DHCP client are different, it is recommended that you install the DHCPv6 server in the underlay so that there is only one instance of the DHCPv6 server serving overlay clients.

### Configuring the DHCPv6 relay agent



DHCP relay is supported by the Aruba 4100i Switch Series but not the Aruba 6000 or 6100 Switch Series.

### Prerequisites

- An enabled layer 3 interface.

### Procedure

1. Enable the DHCPv6 agent with the command `dhcpv6-relay`.
2. Configure one or more IP helper addresses with the command `ipv6 helper-address`. This determines where the DHCPv6 agent forward DHCP requests.
3. If you want to enable DHCP option 79 support to forward client link-layer addresses, use the command `dhcpv6-relay option 79`.
4. Review DHCPv6 relay agent configuration settings with the commands `show dhcpv6-relay` and `show ipv6 helper-address`.

### Example

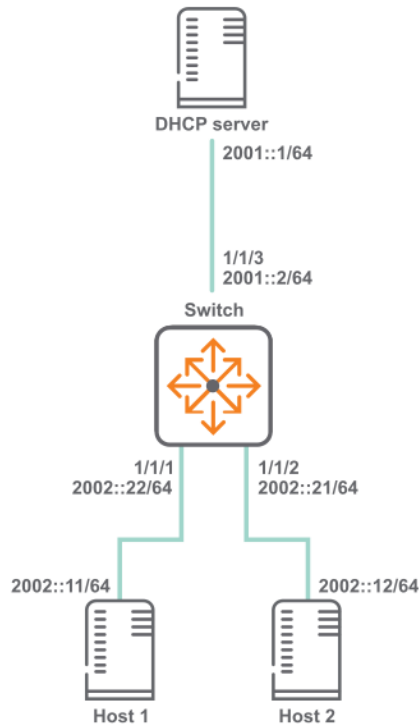
This example creates the following configuration:

- Enables the DHCPv6 relay agent.
- Enables interface **1/1/2** and assigns an IPv6 address to it. (By default, all interfaces are layer 3 and disabled.)
- Defines an IP helper address of **FF01::1:1000** on interface **1/1/2**.
- Enables DHCP option 79.

```
switch(config)# dhcpv6-relay
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
switch(config-if)# ip helper-address FF01::1:1000
switch(config-if)# exit
switch(config)# dhcpv6-relay option 79
```

## DHCPv6 relay scenario 1

In this scenario, DHCP relay on the server enables two hosts to obtain their IP addresses from a DHCP server on a different subnet. The physical topology of the network looks like this:



## Procedure

1. Enable DHCP relay.

```
switch# config
switch(config)# dhcpv6-relay
```

2. Define an IPv6 helper address on interfaces 1/1/1 and 1/1/2 .

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 address 2002::22/64
switch(config-if)# ipv6 helper-address 2001::1
switch(config-if)# interface 1/1/2
switch(config-if)# ipv6 address 2002::21/64
switch(config-if)# ipv6 helper-address 2001::1
switch(config-if)# quit
```

3. Verify DHCP relay configuration.

```
switch# show dhcpv6-relay
DHCPv6 Relay Agent : Enabled
Option 79 : Disabled
switch# show ipv6 helper-address
```

|                     |             |
|---------------------|-------------|
| Interface: 1/1/1    |             |
| IPv6 Helper Address | Egress Port |
| -----               | -----       |
| 2001::1             | 1/1/3       |
|                     |             |
| Interface: 1/1/2    |             |
| IPv6 Helper Address | Egress Port |
| -----               | -----       |
| 2001::1             | 1/1/3       |

## DHCP relay (IPv6) commands

## dhcpv6-relay

dhcpv6-relay  
no dhcpv6-relay

### Description

Enables DHCPv6 relay support. DHCPv6 relay is disabled by default.  
DHCP relay is not supported on the management interface  
The `no` form of this command disables DHCP relay support.

### Examples

Enables DHCPv6 relay support.

```
switch(config)# dhcpv6-relay
```

Removes DHCPv6 relay support.

```
switch(config)# no dhcpv6-relay
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv6-relay option 79

dhcpv6-relay option 79  
no dhcpv6-relay option 79

### Description

Enables support for DHCP relay option 79. When enabled, the DHCPv6 relay agent forwards the link-layer address of the client. This option is disabled by default.  
The `no` form of this command disables support for DHCP relay option 79.

### Examples

Enables DHCP option 79 support.

```
switch(config)# dhcpv6-relay option 79
```

Disables DHCP option 79 support.



```
switch(config)# no dhcpv6-relay option 79
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## ipv6 helper-address

```
ipv6 helper-address unicast <UNICAST-IPV6-ADDR>
no ipv6 helper-address unicast <UNICAST-IPV6-ADDR>
ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>} egress <PORT-NUM>
no ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>} egress <PORT-NUM>
```

## Description

Defines the address of a remote DHCPv6 server or DHCPv6 relay agent. Up to eight addresses can be defined. The DHCPv6 agent forwards DHCPv6 client requests to all defined servers.

Not supported on the OOBM interface.

The `no` form of this command removes an IP helper address.

| Parameter             | Description                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <UNICAST-IPV6-ADDR>   | Specifies the unicast helper IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.                                                                                   |
| <MULTICAST-IPV6-ADDR> | Specifies the multicast helper IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.                                                                                 |
| all-dhcp-servers      | Specifies all the DHCP server IPv6 addresses for the interface.                                                                                                                                                                  |
| egress <PORT-NUM>     | Specifies the port number on which DHCPv6 service requests are relayed to a multicast destination. The egress port must be different than the one on which the multicast helper address is configured. Format: member/slot/port. |
| vrf <VRF-NAME>        | Specifies the name of the VRF from which the specified protocol sets its source IP address.                                                                                                                                      |

## Examples

Defining a multicast IPv6 helper address of **2001:DB8::1** on port **1/1/2**:

```
switch(config-if)# ipv6 helper-address multicast 2001:DB8:0:0:0:0:1 egress 1/1/2
```

Removing the IP helper address of **2001:DB8::1** on port **1/1/2**:

```
switch(config-if)# no ipv6 helper-address multicast 2001:DB8:0:0:0:0:1 egress 1/1/2
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config-if       | Administrators or local user group members with execution rights for this command. |

## show dhcpv6-relay

```
show dhcpv6-relay
```

## Description

Shows DHCP relay configuration settings.

## Example

```
switch# show dhcpv6-relay
 DHCPv6 Relay Agent : Enabled
 Option 79 : Enabled
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                                                                                                              |
|-----------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Manager (#)     | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 helper-address

```
show ipv6 helper-address [interface <INTERFACE-ID>]
```

## Description

Shows the helper IP addresses defined for all interfaces or a specific interface.

| Parameter                | Description                                       |
|--------------------------|---------------------------------------------------|
| interface <INTERFACE-ID> | Specifies an interface. Format: member/slot/port. |

## Examples

```
switch# show ipv6 helper-address

Interface: 1/1/1
IPv6 Helper Address Egress Port

2001:db8:0:1:: -
FF01::1:1000 1/1/2

Interface: 1/1/2
IPv6 Helper Address Egress Port

2001:db8:0:1:: -

switch# show ipv6 helper-address interface 1/1/1

Interface: 1/1/1
IPv6 Helper Address Egress Port

2001:db8:0:1:: -
FF01::1:1000 1/1/2
```

```
switch# show ipv6 helper-address interface 1/1/1

Interface: 1/1/1
IPv6 Helper Address Egress Port

2001:db8:0:1:: -
FF01::1:1000 1/1/2
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                                                                                                              |
|-----------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Manager (#)     | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## http-proxy

```
http-proxy {<FQDN | IPV4-ADDR>} [vrf <VRF-NAME>]
no http-proxy [<FQDN | IPV4-ADDR>] [vrf <VRF-NAME>]
```

## Description

Specifies HTTP proxy location and VRF.

The `no` form of this command removes a specified HTTP proxy location.

| Parameter                      | Description                                     |
|--------------------------------|-------------------------------------------------|
| <code>&lt;FQDN&gt;</code>      | Specifies FQDN for HTTP proxy location.         |
| <code>&lt;IPv4-ADDR&gt;</code> | Specifies IPv4 address for HTTP proxy location. |
| <code>&lt;VRF-NAME&gt;</code>  | Specifies VRF for HTTP proxy.                   |



A FQDN or IPv4 address are optional in the `no` form of the command.

## Usage

- HTTP proxy location can be configured using the CLI/REST interface or auto-configured through the DHCP server connected to the switch.
- There are three sources for HTTP proxy location:
  - User configured HTTP proxy via CLI or REST interface.
  - DHCP options received via management/OOBM port.
  - DHCP options received via VLAN 1 on supported switch platforms.
- Operational configuration for HTTP proxy location is determined by the source with the highest priority. Source priority:
  1. User configured.
  2. DHCP options received via management/OOBM port.
  3. DHCP options received via VLAN 1.



- HTTP proxy location can only be a FQDN or an IPv4 address.
- When HTTP proxy location and VRF are configured, they override any existing HTTP proxy location and VRF.
- If this command is executed without the VRF parameter, the default VRF will be used.
- Port number may need to be specified at the end of the IP address for FQDN to connect via HTTP proxy.
  - For example, 8088 is the TCP port number: `http-proxy 192.168.248.248:8088`

## Examples

Specifying a FQDN for HTTP proxy location and `MGMT` VRF:

```
switch(config)# http-proxy http-proxy.aruba.com vrf mgmt
```

Removing HTTP proxy location

```
switch(config)# no http-proxy
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context     | Authority                                                                          |
|---------------|---------------------|------------------------------------------------------------------------------------|
| All platforms | <code>config</code> | Administrators or local user group members with execution rights for this command. |



DHCP snooping is supported by the Aruba 4100i Switch Series but not the Aruba 6000 or 6100 Switch Series.

### Overview

DHCP is a protocol used by DHCP servers in IP networks to dynamically allocate network configuration data to client devices (DHCP clients). Possible network configuration data includes user IP address, subnet mask, default gateway IP address, DNS server IP address, and lease duration. The DHCP protocol enables DHCP clients to be dynamically configured with such network configuration data without any manual setup process.

DHCP snooping is a security feature that helps avoid problems caused by an unauthorized DHCP server on the network that provides invalid configuration data to DHCP clients. A user without malicious intent may cause this problem by unknowingly adding to the network a switch or other device that includes a DHCP server enabled by default. In some cases, a user with malicious intent adds a DHCP server to the network as part of their Denial of Service or Man in the Middle attack.

DHCP snooping helps prevent such problems by distinguishing between trusted ports connected to legitimate DHCP servers, and untrusted ports connected to general users. DHCP packets are forwarded between trusted ports without inspection. DHCP packets received on other switch ports are inspected before being forwarded. DHCP Packets from untrusted sources are dropped.

In addition, in support of the separate IP source lockdown feature, DHCP snooping also dynamically collects client information (VLAN, IPv4 address, MAC address, interface), adding the information to the switch IP binding database. Alternatively, also in support of IP lockdown, the IP binding database can be statically updated using the `ipv4 source-binding` or `ipv6 source-binding` commands. Statically configured IP binding information supersedes any dynamically collected information for the same client.

### DHCP server interoperation

DHCP server may not be configured with DHCP snooping.

### DHCPv4 snooping conditions for dropping DHCPv4 packets

Applies only to DHCPv4 snooping.

| Packet types that are dropped | Conditions for dropping the packets                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHCPOFFER, DHCPACK, DHCPNACK  | <ul style="list-style-type: none"><li>■ A packet from a DHCP server is received on an untrusted port.</li><li>■ The switch is configured with a list of authorized DHCP server addresses and a packet is received from a DHCP server on a trusted port with a source IP address that is not in the list of authorized DHCP server addresses.</li></ul> |
| DHCPRELEASE, DHCPDECLINE      | A broadcast packet that has a MAC address in the DHCP binding database, but the port in the DHCP binding database does not match the port on which                                                                                                                                                                                                     |

| Packet types that are dropped | Conditions for dropping the packets                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | the packet is received.                                                                                                                                                                                                                                                                                                                                          |
| All DHCP packet types         | <ul style="list-style-type: none"> <li>When enabled (the default) a DHCP packet received on an untrusted port in which the DHCP client hardware MAC address does not match the source MAC address in the packet.</li> <li>When enabled (the default), a DHCP packet containing DHCP relay information (option 82) is received from an untrusted port.</li> </ul> |

## DHCPv4 snooping commands

### clear dhcpv4-snooping binding

```
clear dhcpv4-snooping binding {all | ip <IPV4-ADDR> vlan <VLAN-ID> | port <PORT-NUM> | vlan <VLAN-ID>}
```

#### Description

Clears DHCPv4 snooping binding entries.

| Parameter                     | Description                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------|
| all                           | Specifies that all DHCPv4 binding information is to be cleared.                                |
| ip <IPV4-ADDR> vlan <VLAN-ID> | Specifies the IPv4 address and VLAN for which all DHCPv4 binding information is to be cleared. |
| port <PORT-NUM>               | Specifies the port number for which all DHCPv4 binding information is to be cleared.           |
| vlan <VLAN-ID>                | Specifies the VLAN for which all DHCPv4 binding information is to be cleared.                  |

#### Examples

Clearing all DHCPv4 binding information for IP address 192.168.2.4 and VLAN 5:

```
switch(config)# clear dhcpv4-snooping binding ip 192.168.2.4 vlan 5
```

Clearing all DHCPv4 binding information for port 1/1/1:

```
switch(config)# clear dhcpv4-snooping binding port 1/1/1
```

Clearing all DHCPv4 binding information for VLAN 10:

```
switch(config)# clear dhcpv4-snooping binding vlan 10
```

Clearing all DHCPv4 binding information:

```
switch(config)# clear dhcpv4-snooping binding all
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear dhcpv4-snooping statistics

```
clear dhcpv4-snooping statistics
```

### Description

Clears all DHCPv4 snooping statistics.

### Examples

Clear all DHCPv4 snooping statistics:

```
switch# clear dhcpv4-snooping statistics
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## dhcpv4-snooping

```
dhcpv4-snooping
no dhcpv4-snooping
```

### Description

Enables DHCPv4 snooping. DHCPv4 snooping is disabled by default. DHCP snooping is not supported on the management interface.



The no form of the command disables DHCPv4 snooping, flushing all the IP bindings learned since DHCPv4 snooping was enabled.

## Examples

Enabling DHCPv4 snooping:

```
switch(config)# dhcpv4-snooping
```

Disabling DHCPv4 snooping:

```
switch(config)# no dhcpv4-snooping
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv4-snooping (in config-vlan context)

dhcpv4-snooping  
no dhcpv4-snooping

## Description

Enables DHCPv4 snooping in the `config-vlan` context. DHCPv4 snooping is disabled by default for all VLANs.

The no form of the command disables DHCPv4 snooping on the specified VLAN, flushing all the IP bindings learned for this VLAN since DHCPv4 snooping was enabled for this VLAN.

## Examples

Enabling DHCPv4 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# dhcpv4-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling DHCPv4 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no dhcpv4-snooping
```

```
switch(config-vlan-100) # exit
switch(config) #
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config-vlan     | Administrators or local user group members with execution rights for this command. |

## dhcpv4-snooping allow-overwrite-binding

```
dhcpv4-snooping allow-overwrite-binding
no dhcpv4-snooping allow-overwrite-binding
```

### Description

Allows binding to be overwritten for the same IP address. When enabled, and a DHCP server offers a host an IP address that is already bound to an existing host in the binding table, the existing binding is overwritten for the new host if the new host is successfully able to acquire the same IP address. This overwriting is disabled by default, causing the DHCP server offers to be dropped.

The no form of the command disables DHCPv4 snooping overwrite binding.

### Examples

Enabling DHCPv4 snooping overwrite binding:

```
switch(config) # dhcpv4-snooping allow-overwrite-binding
```

Disabling DHCPv4 snooping overwrite binding:

```
switch(config) # no dhcpv4-snooping allow-overwrite-binding
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv4-snooping authorized-server

```
dhcpv4-snooping authorized-server <IPv4-ADDR> [vrf <VRF-NAME>]
no dhcpv4-snooping authorized-server <IPv4-ADDR> [vrf <VRF-NAME>]
```

### Description

Adds an authorized (trusted) DHCP server to a list of authorized servers for use by DHCPv4 snooping. This command can be issued multiple times, adding a maximum of 20 authorized servers per VRF. By default, with an empty list of authorized servers, all DHCP servers are considered to be trusted for DHCPv4 snooping purposes.



The `mgmt` VRF cannot be used with this command.

The `no` form of this command deletes the specified DHCP server from the authorized list.

| Parameter      | Description                                              |
|----------------|----------------------------------------------------------|
| <IPv4-ADDR>    | Specifies the IPv4 address of the trusted DHCPv4 server. |
| vrf <VRF-NAME> | Specifies the VRF name.                                  |

### Usage

For authorized server lookup, the VRF is derived from the Switch Virtual Interface (SVI) configured for the incoming VLAN. If the SVI is not configured, the `default` VRF is assumed.

### Examples

Adding DHCP servers 192.168.2.2, 192.168.2.3, and 192.168.2.10 to the authorized server list:

```
switch(config)# dhcpv4-snooping authorized-server 192.168.2.2
switch(config)# dhcpv4-snooping authorized-server 192.168.2.3 vrf default
switch(config)# dhcpv4-snooping authorized-server 192.168.2.10 vrf default
```

Removing DHCP server 192.168.2.3 from the authorized server list:

```
switch(config)# no dhcpv4-snooping authorized-server 192.168.2.3 vrf default
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv4-snooping external-storage

```
dhcpv4-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
no dhcpv4-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
```

## Description

Configures external storage to be used for backing up IP bindings (used by DHCPv4 snooping) to a file. When configured, the switch stores all the IP bindings in an external storage file so that they are retained after the switch restarts. When the switch restarts, it reads the IP bindings from the configured external storage file to populate its local cache.



When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IP bindings in an external storage file.

| Parameter         | Description                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| volume <VOL-NAME> | Specifies the name of the existing external storage volume where the IP bindings file will be saved. Before running the <code>dhcpv4-snooping external-storage volume</code> command, first create the external storage volume using command <code>external-storage &lt;VOLUME-NAME&gt;</code> . See <i>External storage commands</i> in the <i>Command-Line Interface Guide</i> . |
| file <FILE-NAME>  | Specifies the file name to use for storing IP bindings. Maximum 255 characters.                                                                                                                                                                                                                                                                                                    |

Configuring IP bindings storage in file `dsnoop_ipbindings` on existing volume `dhcp_snoop`:

```
switch(config)# dhcpv4-snooping external-storage volume dhcp_snoop file dsnoop_
ipbindings
```

Disabling external storage:

```
switch(config)# no dhcpv4-snooping external-storage volume dhcp_snoop
```

Disabling external storage when flash storage is also configured (note the message indicating that flash storage will be used):

```
switch(config)# no dhcpv4-snooping external-storage volume dhcp_snoop
DHCPv4-Snooping will use flash storage to store IP Binding database
switch(config)#
```

## Command History

| Release          | Modification                     |
|------------------|----------------------------------|
| 10.08            | Added flash storage information. |
| 10.07 or earlier | --                               |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv4-snooping flash-storage

```
dhcpv4-snooping flash-storage [delay <DELAY>]
no dhcpv4-snooping flash-storage [delay <DELAY>]
```

### Description

Configures switch flash storage to be used for backing up client IP bindings (used by DHCPv4 snooping). When flash storage is configured (and external storage is not already configured for this purpose), the switch stores the IP bindings in switch flash storage. When the switch restarts, it reads the IP bindings from the switch flash storage to populate its local cache.

Writing the IP bindings to flash storage only occurs after the configured delay and if there has been a change in client IP bindings. Writing is skipped when client IP bindings have not changed since the previous write.

Omitting `delay <DELAY>` sets the default delay of 900 seconds.



To reduce switch flash aging it is recommended that you use external storage (command `dhcpv4-snooping external-storage`) to backup DHCP snooping IP bindings. Alternatively, consider configuring flash storage with a substantial delay between writes.



When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The `no` form of this command disables the saving of IP bindings in flash storage.

| Parameter                        | Description                                                                                                             |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <code>delay &lt;DELAY&gt;</code> | Specifies the delay in seconds between writes (when necessary) to the flash storage, Default: 900. Range: 300 to 86400. |

### Examples

Configuring switch flash storage for DHCP snooping IP binding storage with a write delay of 1200 seconds:

```
switch(config)# dhcpv4-snooping flash-storage delay 1200
Warning: Using flash storage reduces switch lifetime. It is recommended to use an
external-storage.
Do you want to continue (y/n)? y
switch(config)#
```

Unconfiguring usage of switch flash storage for IP bindings :

```
switch(config)# no dhcpv4-snooping flash-storage
```

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.08   | Command introduced. |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv4-snooping max-bindings

```
dhcpv4-snooping max-bindings <MAX-BINDINGS>
no dhcpv4-snooping max-bindings <MAX-BINDINGS>
```

### Description

Sets the maximum number of DHCP bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max bindings applies.

The no form of the command reverts max bindings for the selected interface to its default.

| Parameter      | Description                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <MAX-BINDINGS> | Specifies the maximum number of DHCP bindings. You can use the show capacities command to see the maximum available for your switch model. |

## Examples

Set the DHCP max bindings to 512 on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcpv4-snooping max-bindings 512
switch(config-if)# exit
switch(config)#
```

Revert DHCP max bindings to its default on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcpv4-snooping max-bindings 512
switch(config-if)# exit
switch(config)#
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config-if       | Administrators or local user group members with execution rights for this command. |

## dhcpv4-snooping option 82

```

dhcpv4-snooping option 82 [remote-id {mac | subnet-ip | mgmt-ip}]
 [untrusted-policy {drop | keep | replace}]
no dhcpv4-snooping option 82 [remote-id {mac | subnet-ip | mgmt-ip}]
 [untrusted-policy {drop | keep | replace}]

```

## Description

Configures the addition of option 82 DHCP relay information to DHCP client packets that are being forwarded on trusted ports. DHCP relay is enabled by default.

In the switch default state and when this command is entered without parameters (`dhcpv4-snooping option 82`), this default configuration is used:

```
dhcpv4-snooping option 82 remote-id mac untrusted-policy drop
```

When `remote-id` is omitted, its default (`mac`) is used. When `untrusted-policy` is omitted, its default (`drop`) is used.

The no form of this command disables DHCPv4 snooping option 82.

| Parameter                     | Description                                                                                                                                                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>remote-id</code>        | Specifies what address to use as the remote ID for the <code>replace</code> option of <code>untrusted-policy</code> . Specify one of these address types:                                                     |
| <code>mac</code>              | The default. Uses the switch MAC address as the remote ID.                                                                                                                                                    |
| <code>subnet-ip</code>        | Uses the IP address of the client VLAN as the remote ID.                                                                                                                                                      |
| <code>mgmt-ip</code>          | Uses the management interface IP address as the remote ID.                                                                                                                                                    |
| <code>untrusted-policy</code> | Specifies what action to take for DHCP packets (with option 82) that are received on untrusted ports. Specify one of these actions:                                                                           |
| <code>drop</code>             | The default. Drop DHCP packets (with option 82) without forwarding them.                                                                                                                                      |
| <code>keep</code>             | Forward DHCP packets (with option 82).                                                                                                                                                                        |
| <code>replace</code>          | Replace the option 82 information in the DHCP packets with whatever is set for <code>remote-id</code> (one of: <code>mac</code> , <code>subnet-ip</code> , or <code>mgmt-ip</code> ) and forward the packets. |

## Examples

Configuring DHCPv4 snooping option 82 with the keep action:

```
switch(config)# dhcpv4-snooping option 82 untrusted-policy keep
```

Configuring DHCPv4 snooping option 82 with `mgmt-ip` as the `remote-id` and the `replace` action:

```
switch(config)# dhcpv4-snooping option 82 remote-id mgmt-ip untrusted-policy replace
```

Disabling DHCPv4 snooping option 82:

```
switch(config)# no dhcpv4-snooping option 82 untrusted-policy keep
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv4-snooping trust

dhcpv4-snooping trust  
no dhcpv4-snooping trust

### Description

Enables DHCPv4 snooping trust on the selected port. Only server packets received on trusted ports are forwarded. All the ports are untrusted by default.

The `no` form of the command disables DHCPv4 snooping trust on the selected port.

### Examples

Enabling DHCPv4 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcpv4-snooping trust
switch(config-if)# exit
switch(config)#
```

Disabling DHCPv4 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcpv4-snooping trust
switch(config-if)# exit
switch(config)#
```

## Command History



| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config-if       | Administrators or local user group members with execution rights for this command. |

## dhcpv4-snooping verify mac

dhcpv4-snooping verify mac  
no dhcpv4-snooping verify mac

### Description

This command enables verification of the hardware address field in DHCP client packets. When enabled, the DHCP client hardware address field and the source MAC address must be the same for packets received on untrusted ports or else the packet is dropped. This DHCP snooping MAC verification is enabled by default. The no form of the command disables DHCPv4 snooping MAC verification.

### Examples

Enabling DHCPv4 snooping MAC verification:

```
switch(config)# dhcpv4-snooping verify mac
```

Disabling DHCPv4 snooping MAC verification:

```
switch(config)# no dhcpv4-snooping verify mac
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## show dhcpv4-snooping

show dhcpv4-snooping

### Description

Shows the DHCPv4 snooping configuration.

## Examples

Showing the DHCPv4 snooping configuration:

```
switch(config)# show dhcpv4-snooping

DHCPv4-Snooping Information

 DHCPv4-Snooping : Yes Verify MAC Address : Yes
 Allow Overwrite Binding : Yes Enabled VLANs : 1,5,7,100-110

Option 82 Configurations

 Untrusted Policy : replace Insertion : Yes
 Option 82 Remote-id : mac

External Storage Information

 Volume Name : ipbinding
 File Name : ipv4Bindings
 Inactive Since : 01:23:20 09/10/2021
 Error : File Write Failure

Flash Storage Information

 File Write Delay : 300 seconds

Active Storage : External

Authorized Server Configurations

VRF Authorized Servers

default 1.1.10.3
default 10.10.10.1
default 10.10.10.56
default 200.10.10.3
green 1.1.10.3
green 1.10.10.3
green 10.10.100.3
red 192.168.122.53
red 192.168.122.121

Port Information

 Port Trust Max Bindings Static Bindings Dynamic Bindings

 1/1/2 Yes 5000 50 0
 1/1/3 Yes 8192 0 0
 1/1/5 Yes 8192 0 22
 1/1/16 No 100 0 0
 10/10/10 No 8100 320 200
 lag120 No 512 0 0

VLAN Information

 VLAN Guard Policy Bind Disable Guard Disable

 1 POL1 True True
```

|   |      |       |       |
|---|------|-------|-------|
| 5 |      | False | False |
| 7 | POL2 | True  | True  |

## Command History

| Release          | Modification                                    |
|------------------|-------------------------------------------------|
| 10.08            | Updated example with flash storage information. |
| 10.07 or earlier | --                                              |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show dhcpv4-snooping binding

show dhcpv4-snooping binding

### Description

Shows the DHCPv4 snooping binding configuration.

### Examples

Showing the DHCPv4 snooping binding configuration:

```
switch(config)# show dhcpv4-snooping binding
```

| MacAddress        | IP       | VLAN | Interface | Time-Left |
|-------------------|----------|------|-----------|-----------|
| aa:b1:c1:dd:ee:ff | 10.2.3.4 | 1    | 1/1/2     | 582       |
| aa:b2:c2:dd:ee:ff | 10.2.3.5 | 1    | 1/1/2     | 584       |

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show dhcpv4-snooping statistics

```
show dhcpv4-snooping statistics
```

## Description

Shows the DHCPv4 snooping statistics.

## Examples

Showing the DHCPv4 snooping statistics:

```
switch(config)# show dhcpv4-snooping statistics
```

| Packet-Type | Action  | Reason                        | Count |
|-------------|---------|-------------------------------|-------|
| server      | forward | from trusted port             | 5425  |
| client      | forward | to trusted port               | 3895  |
| server      | drop    | received on untrusted port    | 117   |
| server      | drop    | unauthorized server           | 214   |
| client      | drop    | destination on untrusted port | 78    |
| client      | drop    | untrusted option 82 field     | 85    |
| client      | drop    | bad DHCP release request      | 0     |
| client      | drop    | failed verify MAC check       | 5     |
| client      | drop    | failed on max-binding limit   | 15    |

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## DHCPv6 snooping commands

### clear dhcpv6-snooping binding

```
clear dhcpv6-snooping binding {all | ip <IPV6-ADDR> vlan <VLAN-ID> | interface <IFNAME> |
vlan <VLAN-ID>}
```

## Description

Clears DHCPv6 snooping binding entries.

| Parameter                     | Description                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------|
| all                           | Specifies that all DHCPv6 binding information is to be cleared.                                |
| ip <IPV6-ADDR> vlan <VLAN-ID> | Specifies the IPv6 address and VLAN for which all DHCPv6 binding information is to be cleared. |

| Parameter          | Description                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------|
| interface <IFNAME> | Specifies the interface for which all DHCPv6 binding information is to be cleared.              |
| vlan <VLAN-ID>     | Specifies the VLAN for which all DHCPv6 binding information is to be cleared. Range: 1 to 4094. |

## Examples

Clearing all DHCPv6 binding information for 5000::1 vlan 1:

```
switch(config)# clear dhcpv6-snooping binding ip 5000::1 vlan 1
```

Clearing all DHCPv6 binding information for interface 1/1/10:

```
switch(config)# clear dhcpv6-snooping binding interface 1/1/10
```

Clearing all DHCPv6 binding information for VLAN 10:

```
switch(config)# clear dhcpv6-snooping binding vlan 10
```

Clearing all DHCPv6 binding information:

```
switch(config)# clear dhcpv6-snooping binding all
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## clear dhcpv6-snooping statistics

```
clear dhcpv6-snooping statistics
```

### Description

Clears all DHCPv6 snooping statistics.

### Examples

Clear all DHCPv6 snooping statistics:

```
switch# clear dhcpv6-snooping statistics
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## dhcpv6-snooping

dhcpv6-snooping  
no dhcpv6-snooping

### Description

Enables DHCPv6 snooping. DHCPv6 snooping is disabled by default. DHCPv6 snooping is not supported on the management interface.

The no form of the command disables DHCPv6 snooping, flushing all the IP bindings learned since DHCPv6 snooping was enabled.

### Examples

Enabling DHCPv6 snooping:

```
switch(config)# dhcpv6-snooping
```

Disabling DHCPv6 snooping:

```
switch(config)# no dhcpv6-snooping
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv6-snooping (in config-vlan context)

dhcpv6-snooping  
no dhcpv6-snooping

### Description

Enables DHCPv6 snooping in the `config-vlan` context. DHCPv6 snooping is disabled by default for all VLANs.

The `no` form of the command disables DHCPv6 snooping on the specified VLAN, flushing all the IPv6 bindings learned for this VLAN since DHCPv6 snooping was enabled for this VLAN.

### Examples

Enabling DHCPv6 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# dhcpv6-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling DHCPv6 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no dhcpv6-snooping
switch(config-vlan-100)# exit
switch(config)#
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config-vlan     | Administrators or local user group members with execution rights for this command. |

## dhcpv6-snooping authorized-server

dhcpv6-snooping authorized-server <IPv6-ADDR> [vrf <VRF-NAME>]  
no dhcpv6-snooping authorized-server <IPv6-ADDR> [vrf <VRF-NAME>]

### Description

Adds an authorized (trusted) DHCPv6 server to a list of authorized servers for use by DHCPv6 snooping. This command can be issued multiple times, adding a maximum of 20 authorized servers per VRF. By default, with an empty list of authorized servers, all DHCPv6 servers are considered to be trusted for DHCPv6 snooping purposes.



The `mgmt` VRF cannot be used with this command.



Configure the link local IPv6 address instead of global IPv6 address of the DHCPv6 server as the authorized-server. For example:

```
switch(config)# dhcpv6-snooping authorized-server fe80::2ca4:fa40:d4cd:bc2f
```

The no form of this command deletes the specified DHCPv6 server from the authorized list.

| Parameter      | Description                                              |
|----------------|----------------------------------------------------------|
| <IPv6-ADDR>    | Specifies the IPv6 address of the trusted DHCPv6 server. |
| vrf <VRF-NAME> | Specifies the VRF name.                                  |

## Usage

For authorized server lookup, the VRF is derived from the Switch Virtual Interface (SVI) configured for the incoming VLAN. If the SVI is not configured, the `default` VRF is assumed.

## Examples

Adding DHCP servers ABCD:5ACD::2000, and ABCD:5ACD::2010 to the authorized server list:

```
switch(config)# dhcpv6-snooping authorized-server ABCD:5ACD::2000 vrf default
switch(config)# dhcpv6-snooping authorized-server ABCD:5ACD::2010 vrf default
```

Removing DHCP server ABCD:5ACD::2000 from the authorized server list:

```
switch(config)# no dhcpv6-snooping authorized-server ABCD:5ACD::2000 vrf default
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv6-snooping external-storage

```
dhcpv6-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
no dhcpv6-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
```

## Description



Configures external storage to be used for backing up IPv6 bindings (used by DHCPv6 snooping) to a file. When configured, the switch stores all the IP bindings in an external storage file so that they are retained after the switch restarts. When the switch restarts, it reads the IPv6 bindings from the configured external storage file to populate its local cache.



When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IPv6 bindings in an external storage file.

| Parameter                            | Description                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>volume &lt;VOL-NAME&gt;</code> | Specifies the name of the existing external storage volume where the IPv6 bindings file will be saved. Before running the <code>dhcpv6-snooping external-storage volume</code> command, first create the external storage volume using command <code>external-storage &lt;VOLUME-NAME&gt;</code> . See <i>External storage commands</i> in the <i>Command-Line Interface Guide</i> . |
| <code>file &lt;FILE-NAME&gt;</code>  | Specifies the file name to use for storing IPv6 bindings. Maximum 255 characters.                                                                                                                                                                                                                                                                                                    |

## Examples

Configuring IPv6 bindings storage in file `ipv6Bindings` on existing volume `dhcp_snoop`:

```
switch(config)# dhcpv6-snooping external-storage volume dhcp_snoop file ipv6Bindings
```

Disabling external storage:

```
switch(config)# no dhcpv6-snooping external-storage volume dhcp_snoop
```

Disabling external storage when flash storage is also configured (note the message indicating that flash storage will be used):

```
switch(config)# no dhcpv6-snooping external-storage volume dhcp_snoop
DHCPv6-Snooping will use flash storage to store IP Binding database
switch(config)#
```

## Command History

| Release          | Modification                     |
|------------------|----------------------------------|
| 10.08            | Added flash storage information. |
| 10.07 or earlier | --                               |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv6-snooping flash-storage

```
dhcpv6-snooping flash-storage [delay <DELAY>]
no dhcpv6-snooping flash-storage [delay <DELAY>]
```

### Description

Configures switch flash storage to be used for backing up client IP bindings (used by DHCPv6 snooping). When flash storage is configured (and external storage is not already configured for this purpose), the switch stores the IP bindings in switch flash storage. When the switch restarts, it reads the IP bindings from the switch flash storage to populate its local cache.

Writing the IP bindings to flash storage only occurs after the configured delay and if there has been a change in client IP bindings. Writing is skipped when client IP bindings have not changed since the previous write.

Omitting `delay <DELAY>` sets the default delay of 900 seconds.



To reduce switch flash aging it is recommended that you use external storage (command `dhcpv6-snooping external-storage`) to backup DHCP snooping IP bindings. Alternatively, consider configuring flash storage with a substantial delay between writes.



When both external storage and flash storage are configured to store DHCP snooping IP bindings, the external storage takes priority, and is used exclusively until it becomes unconfigured, at which time flash storage (if configured) is used. Later, if external storage is configured again, flash storage stops and external storage resumes.

The no form of this command disables the saving of IP bindings in flash storage.

| Parameter                        | Description                                                                                                             |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <code>delay &lt;DELAY&gt;</code> | Specifies the delay in seconds between writes (when necessary) to the flash storage, Default: 900. Range: 300 to 86400. |

### Examples

Configuring switch flash storage for DHCP snooping IP binding storage with a write delay of 1200 seconds:

```
switch(config)# dhcpv6-snooping flash-storage delay 1200
Warning: Using flash storage reduces switch lifetime. It is recommended to use an
external-storage.
Do you want to continue (y/n)? y
switch(config)#
```

Unconfiguring usage of switch flash storage for IP bindings :

```
switch(config)# no dhcpv6-snooping flash-storage
```

## Command History

| Release | Modification        |
|---------|---------------------|
| 10.08   | Command introduced. |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## dhcpv6-snooping max-bindings

```
dhcpv6-snooping max-bindings <MAX-BINDINGS>
no dhcpv6-snooping max-bindings <MAX-BINDINGS>
```

### Description

Sets the maximum number of DHCPv6 bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max bindings applies.

The no form of the command reverts max bindings for the selected interface to its default.

| Parameter      | Description                                                                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <MAX-BINDINGS> | Specifies the maximum number of DHCPv6 bindings. You can use the <code>show capacities</code> command to see the maximum available for your switch model. |

### Examples

Set the DHCPv6 max bindings to 256 on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcpv6-snooping max-bindings 256
switch(config-if)# exit
switch(config)#
```

Revert DHCPv6 max bindings to its default on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcpv6-snooping max-bindings 256
switch(config-if)# exit
switch(config)#
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config-if       | Administrators or local user group members with execution rights for this command. |

## dhcpv6-snooping trust

dhcpv6-snooping trust  
no dhcpv6-snooping trust

### Description

Enables DHCPv6 snooping trust on the selected interface. Only server packets received on trusted interfaces are forwarded. All the interfaces are untrusted by default.

The no form of the command disables DHCPv6 snooping trust on the selected interface.

config-if

### Examples

Enabling DHCPv6 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcpv6-snooping trust
switch(config-if)# exit
switch(config)#
```

Disabling DHCPv6 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcpv6-snooping trust
switch(config-if)# exit
switch(config)#
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
| 4100i     | config          | Administrators or local user group members with execution rights for this command. |

## show dhcpv6-snooping

show dhcpv6-snooping

### Description

Shows the DHCPv6 snooping configuration.

### Examples

Showing the DHCPv6 snooping configuration:

```
switch(config)# show dhcpv6-snooping

DHCPv6-Snooping Information

 DHCPv6-Snooping : Yes Enabled VLANs : 1,5,7,100-110

External Storage Information

 Volume Name : dhcp_snoop
 File Name : ip_binding
 Inactive Since : 01:23:20 09/10/2021
 Error : Failed to write external storage

Flash Storage Information

 File Write Delay : 300 seconds

Active Storage : External

Authorized Server Configurations

VRF Authorized Servers

default 2001:0db8:85a3:0000:0000:8a2e:0370:7334
default 2002::2
default 2004::1
red 2002::1
red 2002::2
red 2002::9
green 5000::1
green 5000::2
green 5000::3
green 5000::7
green 5000::8

Port Information

 Port Trust Max Static Dynamic
 ----- -
 1/1/2 Yes 0 0 0
 1/1/3 Yes 0 3 0
 1/1/5 Yes 0 22 0
 1/1/16 No 256 0 20
 10/10/10 No 256 12 7
 lag120 No 256 3 0

VLAN Information

VLAN Guard Bind Guard
 ----- -
 1 POL1 True True
 5 False False
 7 POL2 True True
```

## Command History

| Release          | Modification                                    |
|------------------|-------------------------------------------------|
| 10.08            | Updated example with flash storage information. |
| 10.07 or earlier | --                                              |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show dhcpv6-snooping binding

show dhcpv6-snooping binding

### Description

Shows the DHCPv6 snooping binding configuration.

### Examples

Showing the DHCPv6 snooping binding configuration:

```
switch# show dhcpv6-snooping binding

IP Binding Information
=====
MAC-ADDRESS IPV6-ADDRESS VLAN INTERFACE TIME-
LEFT

00:50:56:96:e4:cf aaaa:bbbb:cccc:dddd:eeee:1234:5678:abcd 1 1/1/1
584
00:50:56:96:04:4d 1000::3 134 1/1/2
435
00:50:56:96:d8:3d 2000:1000::4 2002 lag123
21234
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# show dhcpv6-snooping statistics

show dhcpv6-snooping statistics

## Description

Shows the DHCPv6 snooping statistics.

## Examples

Showing the DHCPv6 snooping statistics:

```
switch(config)# show dhcpv6-snooping statistics
```

| Packet-Type | Action  | Reason                        | Count |
|-------------|---------|-------------------------------|-------|
| server      | forward | from trusted port             | 12    |
| client      | forward | to trusted port               | 20    |
| server      | drop    | received on untrusted port    | 5     |
| server      | drop    | unauthorized server           | 4     |
| client      | drop    | destination on untrusted port | 2     |
| client      | drop    | bad DHCP release request      | 5     |
| server      | drop    | relay reply on untrusted port | 2     |
| client      | drop    | failed on max-binding limit   | 5     |

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4100i     | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

# Chapter 7

## Internet Control Message Protocol (ICMP)

The Internet Control Message Protocol (ICMP) is a supporting protocol in the Internet protocol suite. The protocol is used by network devices, including routers, to send error messages and operational information. For example, an ICMP message might indicate that a requested service is not available. Another example of an ICMP message might be that a host or router could not be reached.

### ICMP message types

The type field identifies the type of message sent by the host or gateway.

| Type | ICMP messages                                             |
|------|-----------------------------------------------------------|
| 0    | Echo Reply (Ping Reply, used with Type 8, Ping Request)   |
| 3    | Destination Unreachable                                   |
| 4    | Source Quench                                             |
| 5    | Redirect                                                  |
| 8    | Echo Request (Ping Request, used with Type 0, Ping Reply) |
| 9    | Router Advertisement (Used with Type 9)                   |
| 10   | Router Solicitation (Used with Type 10)                   |
| 11   | Time Exceeded                                             |
| 12   | Parameter Problem                                         |
| 13   | Timestamp Request (Used with Type 14)                     |
| 14   | Timestamp Reply (Used with Type 13)                       |
| 15   | Information Request (obsolete) (Used with Type 16)        |
| 16   | Information Reply (obsolete) (Used with Type 15)          |
| 17   | Address Mask Request (Used with Type 17)                  |
| 18   | Address Mask Reply (Used with Type 18)                    |

### When ICMP messages are sent



ICMP messages are sent when one or more of the following scenarios occur:

- A datagram cannot reach its destination.
- The gateway does not have the buffering capacity to forward a datagram.
- The gateway can direct the host to send traffic on a shorter route.

## ICMP redirect messages

ICMP redirect messages are used by routers to notify the hosts on the data link that a better route is available for a particular destination.

## When ICMP redirect messages are sent

The switch is configured to send redirects by default. ICMP redirect messages are sent when one or more of the following scenarios occur:

- The interface on which the packet comes into the router is the same interface on which the packet gets routed out.
- The subnet or network of the source IP address is on the same subnet or network of the next-hop IP address of the routed packet.
- The datagram is not source-routed.
- The destination unicast address is unreachable. In this case, the router generates the ICMP destination unreachable message to inform the source host about the situation.

## ICMP commands

### ip icmp redirect

```
ip icmp redirect
no ip icmp redirect
```

#### Description

Enables the sending of ICMPv4 and ICMPv6 redirect messages to the source host. Enabled by default. The `no` form of this command disables ICMPv4 and ICMPv6 redirect messages to the source host.

#### Examples

Enabling ICMP redirect messages:

```
switch(config)# ip icmp redirect
```

Disabling ICMP redirect messages:

```
switch(config)# no ip icmp redirect
```

#### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## ip icmp throttle

```
ip icmp throttle <PACKET-INTERVAL>
no ip icmp throttle [<PACKET-INTERVAL>]
```

## Description

Used to configure the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

The `no` form of this command disables the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

| Parameter         | Description                                                                            |
|-------------------|----------------------------------------------------------------------------------------|
| <PACKET-INTERVAL> | Specifies the ICMPv4/v6 packet interval in seconds. Default: 1 second. Range: 1-86400. |

## Examples

Enabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# ip icmp throttle 3000
```

Disabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# no ip icmp throttle
```

## Command History

| Release          | Modification                                                                               |
|------------------|--------------------------------------------------------------------------------------------|
| 10.8             | Added the optional <PACKET-INTERVAL> parameter to the <code>no</code> form of the command. |
| 10.07 or earlier | --                                                                                         |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## ip icmp unreachable

```
ip icmp unreachable
no ip icmp unreachable
```

### Description

Enables the sending of ICMPv4 and ICMPv6 destination unreachable messages on the switch to a source host when a specific host is unreachable. The unreachable host address originates from the failed packet. Default setting.

The `no` form of this command disables the sending of ICMPv4 and ICMPv6 destination unreachable messages from the switch to a source host when a specific host is unreachable. This command does not prevent other hosts from sending an ICMP unreachable message.

### Examples

Enabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# ip icmp unreachable
```

Disabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# no ip icmp unreachable
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

The Domain Name System (DNS) is the Internet protocol for mapping a hostname to its IP address. DNS allows users to enter more readily memorable and intuitive hostnames, rather than IP addresses, to identify devices connected to a network. It also allows a host to keep the same hostname even if it changes its IP address.

Hostname resolution can be either static or dynamic.

- In static resolution, a local table is defined on the switch that associates hostnames with their IP addresses. Static tables can be used to speed up the resolution of frequently queried hosts.
- Dynamic resolution requires that the switch query a DNS server located elsewhere on the network. Dynamic name resolution takes more time than static name resolution, but requires far less configuration and management.

## DNS client

The DNS client resolves hostnames to IP addresses for protocols that are running on the switch. When the DNS client receives a request to resolve a hostname, it can do so in one of two ways:

- Forward the request to a DNS name server for resolution.
- Reply to the request without using a DNS name server, by resolving the name using a statically defined table of hostnames and their associated IP addresses.

## Configuring the DNS client

### Procedure

1. Configure one or more DNS name servers with the command `ip dns server`.
2. To resolve DNS requests by appending a domain name to the requests, either configure a single domain name with the command `ip dns domain-name`, or configure a list of up to six domain names with the command `ip dns domain-list`.
3. To use static name resolution for certain hosts, associate an IP address to a host with the command `ip dns host`.
4. Review your DNS configuration settings with the command `show ip dns`.

### Examples

This example creates the following configuration:

- Defines the domain **switch.com** to append to all requests.
- Defines a DNS server with IPv4 address of **1.1.1.1**.
- Defines a static DNS host named **myhost1** with an IPv4 address of **3.3.3.3**.
- DNS client traffic is sent on the default VRF (named **default**).

```

switch(config)# ip dns domain-name switch.com
switch(config)# ip dns server-address 1.1.1.1
switch(config)# ip dns host myhost1 3.3.3.3
switch(config)# exit
switch# show ip dns

```

```

VRF Name : default
Domain Name: switch.com
Name Server(s): 1.1.1.1

```

| Host Name | Address |
|-----------|---------|
| -----     | -----   |
| myhost1   | 3.3.3.3 |
| switch#   |         |

## DNS client commands

### ip dns domain-list

```

ip dns domain-list <DOMAIN-NAME> [vrf default]
no ip dns domain-list <DOMAIN-NAME> [vrf default]

```

#### Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The `no` form of this command removes a domain from the list.

| Parameter                             | Description                                                                                       |
|---------------------------------------|---------------------------------------------------------------------------------------------------|
| <code>list &lt;DOMAIN-NAME&gt;</code> | Specifies a domain name. Up to six domains can be added to the list. Length: 1 to 256 characters. |

#### Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```

switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com

```

This example removes the entry **domain1.com**.

```

switch(config)# no ip dns domain-list domain1.com

```

#### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

#### Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## ip dns domain-name

```
ip dns domain-name <DOMAIN-NAME> [vrf <VRF-NAME>]
no ip dns domain-name <DOMAIN-NAME> [vrf <VRF-NAME>]
```

### Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command `ip dns domain-list`, the domain name defined with this command is ignored.

The `no` form of this command removes the domain name.

| Parameter      | Description                                                                       |
|----------------|-----------------------------------------------------------------------------------|
| <DOMAIN-NAME>  | Specifies the domain name to append to DNS requests. Length: 1 to 256 characters. |
| vrf <VRF-NAME> | Specifies a VRF name. Default: default.                                           |

### Examples

Setting the default domain name to `domain.com`:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name `domain.com`:

```
switch(config)# no ip dns domain-name domain.com
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## ip dns host

```
ip dns host <HOST-NAME> <IP-ADDR> [vrf <VRF-NAME>]
no ip dns host <HOST-NAME> <IP-ADDR> [vrf <VRF-NAME>]
```

### Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The `no` form of this command removes a static IP address associated with a hostname.

| Parameter        | Description                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| host <HOST-NAME> | Specifies the name of a host. Length: 1 to 256 characters.                                                                                                                                          |
| <IP-ADDR>        | Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. |
| vrf <VRF-NAME>   | Specifies a VRF name. Default: default.                                                                                                                                                             |

## Examples

This example defines an IPv4 address of **3.3.3.3** for **host1**.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of **b::5** for **host 1**.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for **host 1** with address **b::5**.

```
switch(config)# no ip dns host host1 b::5
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## ip dns server address

```
ip dns server-address <IP-ADDR> [vrf <VRF-NAME>]
no ip dns server-address <IP-ADDR> [vrf <VRF-NAME>]
```

## Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The `no` form of this command removes a name server from the list.

| Parameter                         | Description                                                                                                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;IP-ADDR&gt;</code>      | Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. |
| <code>vrf &lt;VRF-NAME&gt;</code> | Specifies a VRF name. Default: default.                                                                                                                                                             |

## Examples

This example defines a name server at **1.1.1.1**.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at **a::1**.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at **a::1**.

```
switch(config)# no ip dns server-address a::1
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## show ip dns

```
show ip dns [vrf <VRF-NAME>]
```

## Description

Shows all DNS client configuration settings or the settings for a specific VRF.



| Parameter                         | Description                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------------------|
| <code>vrf &lt;VRF-NAME&gt;</code> | Specifies the VRF for which to show information. If no VRF is defined, the default VRF is used. |

## Examples

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                                                                                                              |
|---------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | Manager (#)     | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

ARP (Address Resolution Protocol) is used to map the network address assigned to a device to its physical address. For example, on an Ethernet network, ARP maps layer 3 IPv4 network addresses to layer 2 MAC addresses. (ARP does not work with IPv6 addresses. Instead, the Neighbor discovery protocol is used.)

ARP operates at layer 2. ARP requests are broadcast to all devices on the local network segment and are not forwarded by routers. ARP is enabled by default and cannot be disabled.

### Proxy ARP

Proxy ARP allows a routing switch to answer ARP requests from devices on one network on behalf of devices on another network. The ARP proxy is aware of the location of the traffic destination, and offers its own MAC address as the final destination.

For example, if Proxy ARP is enabled on a routing switch connected to two subnets (10.10.10.0/24 and 20.20.20.0/24), the routing switch can respond to an ARP request from 10.10.10.69 for the MAC address of the device with IP address 20.20.20.69.

Typically, the host that sent the ARP request then sends its packets to the switch that has the ARP proxy. This switch then forwards the packets to the intended host through a mechanism such as a tunnel.

Proxy ARP is supported on L3 physical and VLAN interfaces. It is disabled by default. To enable proxy ARP, routing must be enabled on the interface.

### Local proxy ARP

Local proxy ARP is a technique by which a device on a given network answers the ARP queries for a host address that is on the same network. It is primarily used to enable layer 3 communication between hosts within a common subnet that are separated by layer 2 boundaries (Example: PVLAN). Local proxy ARP is supported on L3 physical and VLAN interfaces.

Local proxy ARP is disabled by default. Routing must be enabled on the interface to enable local proxy ARP.

### Dynamic ARP Inspection

ARP is used for resolving IP against MAC addresses on a broadcast network segment like the Ethernet and was originally defined by Internet Standard RFC 826. ARP does not support any inherent security mechanism and as such depends on simple datagram exchanges for the resolution, with many of these being broadcast.

Because it is an unreliable and non-secure protocol, ARP is vulnerable to attacks. Some attacks may be targeted toward the networks whereas other attacks may be targeted toward the switch itself. The attacks primarily intend to create denial of service (DoS) for the other entities present in the network.

Most of the attacks are carried out in one of the following three forms:

- Overwhelming the switch control plane with too many ARP packets.
- Overwhelming the switch control plane with too many unresolved data packets.
- Masquerading as a trusted gateway/server by wrongly advertising ARPs.

Several defense mechanisms can be put in place on a switch to protect against attacks:

- Limit the amount of ARP activity allowed from a host or on a port.
- Ensure that all ARP packets are consistent with one or more binding databases, which can be created through various means.
- Enforce integrity checks on the ARP packets to check against different MAC or IP addresses in the Ethernet or IP header and ARP header.

This release implements Dynamic ARP Inspection to enforce DHCP snooping binding on all ARP packets and is limited to the 8400 platform. The feature will be disabled from the code, CLI, and schema by the use of appropriate config flags for other platforms.

Only the following is supported:

- Enabling and disabling of Dynamic ARP Inspection on a VLAN level (it does not have to be SVI).
- Defining the member ports of a VLAN as either trusted or untrusted.
- Only ARP traffic on untrusted ports subjected to checks.
- Routed ports (RoPs) always treated as trusted.
- Listening to the DHCP Bindings table and check every ARP packet to match against the binding.

ARP ACLs are not supported in this release and the DHCP snooping table will be the only source of binding.

## ARP commands

### arp inspection

arp inspection

#### Description

Enables Dynamic ARP Inspection on the current VLAN, forcing all ARP packets from untrusted ports to be subjected to a MAC-IP association check against a binding table.

The `no` form of this command disables Dynamic ARP Inspection on the VLAN.

#### Examples

Enabling dynamic ARP inspection:

```
switch# configure terminal
switch(config)# vlan 1
switch(config-vlan)# arp inspection
```

Disabling dynamic ARP inspection:

```
switch# configure terminal
switch(config)# vlan 1
switch(config-vlan)# no arp inspection
```

#### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context                          | Authority                                                                          |
|---------------|------------------------------------------|------------------------------------------------------------------------------------|
| All platforms | <code>config-vlan-&lt;VLAN-ID&gt;</code> | Administrators or local user group members with execution rights for this command. |

## arp inspection trust

arp inspection trust  
no arp inspection trust

### Description

Configures the interface as a trusted. All interfaces are untrusted by default.

The `no` form of this command returns the interface to the default state (untrusted).

### Example

Setting an interface as trusted:

```
switch(config-if)# arp inspection trust
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context        | Authority                                                                          |
|---------------|------------------------|------------------------------------------------------------------------------------|
| All platforms | <code>config-if</code> | Administrators or local user group members with execution rights for this command. |

## arp ipv4 mac

arp ipv4 <IPV4\_ADDR> mac <MAC\_ADDR>  
no arp ipv4 <IPV4\_ADDR> mac <MAC\_ADDR>

### Description

Specifies a permanent static neighbor entry in the ARP table (for IPv4 neighbors).

The `no` form of this command deletes a permanent static neighbor entry from the ARP table.

| Parameter        | Description                                                                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ipv4 <IPV4-ADDR> | Specifies the IP address of the neighbor or the virtual IP address of the cluster in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. . Range: 4096 to 131072. Default: 131072. |
| mac <MAC-ADDR>   | Specifies the MAC address of the neighbor or the multicast MAC                                                                                                                                   |

| Parameter | Description                                                                                                                      |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
|           | address in IANA format (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072. |

## Example

Configuring a static ARP entry on a interface VLAN 10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# arp ipv4 2.2.2.2 mac 01:00:5e:00:00:01
```

Removing a static ARP entry on interface VLAN10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no arp ipv4 2.2.2.2 mac 01:00:5e:00:00:01
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                          |
|---------------|-----------------------------|------------------------------------------------------------------------------------|
| All platforms | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## clear arp

```
clear arp [port <PORT-ID> | vrf {all-vrfs | <VRF-NAME>}]
```

## Description

Clears IPv4 and IPv6 neighbor entries from the ARP table. If you do not specify any parameters, ARP table entries are cleared for the default VRF.

| Parameter      | Description                                                                              |
|----------------|------------------------------------------------------------------------------------------|
| port <PORT-ID> | Specifies a physical port on the switch. Format: member/slot/port. For example: vlan 2.. |
| all-vrfs       | Selects all VRFs.                                                                        |
| <VRF-NAME>     | Specifies the name of a VRF. Default: default.                                           |

## Examples

Clearing all IPv4 and IPv6 neighbor ARP entries for the default VRF:

```
switch# clear arp
```

Clearing all ARP neighbor entries for a port:

```
switch# clear arp vlan 2
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | Manager (#)     | Administrators or local user group members with execution rights for this command. |

## ip local-proxy-arp

```
ip local-proxy-arp
no ip local-proxy-arp
```

### Description

Enables local proxy ARP on the specified interface. Local proxy ARP is supported on Layer 3 physical interfaces and on VLAN interfaces. To enable local proxy ARP on an interface, routing must be enabled on that interface.

The `no` form of this command disables local proxy ARP on the specified interface.

### Examples

Enabling local proxy ARP on interface **1/1/1**:

```
switch# interface 1/1/1
switch(config-if)# ip local proxy-arp
```

Enabling local proxy ARP on interface VLAN **3**:

```
switch# interface vlan 3
switch(config-if-vlan)# ip local-proxy-arp
```

Disabling local proxy ARP on on interface **1/1/1**.

```
switch# interface 1/1/1
switch(config-if)# no ip local-proxy-arp
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                          |
|---------------|-----------------------------|------------------------------------------------------------------------------------|
| All platforms | config-if<br>config-if-vlan | Administrators or local user group members with execution rights for this command. |

## ipv6 neighbor mac

```
ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
no ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
```

## Description

Specifies a permanent static neighbor entry in the ARP table (for IPv6 neighbors).

The `no` form of this command deletes a permanent static neighbor entry from the ARP table.

| Parameter       | Description                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <IPV6-ADDR>>    | Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072. |
| mac <MAC-ADDR>> | Specifies the MAC address of the neighbor (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.                    |

## Example

Creates a static ARP entry on interface **vlan 2**.

```
switch(config)# interface vlan 2
switch(config-if-vlan)# arp ipv6 neighbor 2001:0db8:85a3::8a2e:0370:7334 mac
00:50:56:96:df:c8
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config-if       | Administrators or local user group members with execution rights for this command. |

## ip proxy-arp

```
ip proxy-arp
no ip proxy-arp
```

### Description

Enables proxy ARP for the specified Layer 3 interface. Proxy ARP is supported on Layer 3 physical interfaces, LAG interfaces, and VLAN interfaces. It is disabled by default. To enable proxy ARP on an interface, routing must be enabled on that interface.

The `no` form of this command disables proxy ARP for the specified interface.

### Examples

Enabling proxy ARP on interface **1/1/1**:

```
switch# interface 1/1/1
switch(config-if)# ip proxy-arp
```

Enabling proxy ARP on VLAN **3**:

```
switch# interface vlan 3
switch(config-if-vlan)# ip proxy-arp
```

Enabling proxy ARP on a LAG **11**:

```
switch(config)# int lag 11
switch(config-lag-if)# ip proxy-arp
```

Disabling proxy ARP on interface 1/1/1:

```
switch# interface 1/1/1
switch(config-if)# no ip proxy-arp
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms     | Command context                                | Authority                                                                          |
|---------------|------------------------------------------------|------------------------------------------------------------------------------------|
| All platforms | config-if<br>config-if-vlan<br>config-lag-vlan | Administrators or local user group members with execution rights for this command. |

## show arp

```
show arp
```

### Description



Shows the entries in the ARP (Address Resolution Protocol) table.

## Usage

This command displays information about ARP entries, including the IP address, MAC address, port, and state.

When no parameters are specified, the `show arp` command shows all ARP entries for the default VRF (Virtual Router Forwarding) instance.

## Examples

```
switch# show arp
```

| IPv4 Address | MAC               | Port   | Physical Port |           |
|--------------|-------------------|--------|---------------|-----------|
| 192.168.1.2  | 00:50:56:96:7b:e0 | vlan10 | 1/1/29        | stale     |
| 192.168.1.3  | 00:50:56:96:7b:ac | vlan10 | 1/1/1         | reachable |

Total Number Of ARP Entries Listed- 2.

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | Manager (#)     | Administrators or local user group members with execution rights for this command. |

## show arp inspection interface

```
show arp inspection interface
```

## Description

Displays the current configuration of dynamic ARP inspection on a VLAN or interface.

## Examples

```
switch# show arp inspection interface
```

| Interface | Trust-State |
|-----------|-------------|
| 1/1/1     | Untrusted   |

```
switch# show arp inspection interface vsx-peer
```

```

Interface Trust-State

1/1/1 Untrusted
lag100 Trusted

```

```
switch# show arp inspection interface 1/1/1
```

```

Interface Trust-State

1/1/1 Untrusted

```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                                                                                                              |
|---------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show arp inspection statistics

```
show arp inspection statistics
```

### Description

Displays statistics about forwarded and dropped ARP packets.

### Examples

```
switch# show arp inspection statistics vlan 1-200
```

```

VLAN Name Forwarded Dropped

1 DEFAULT_VLAN_1 0 0

```

```
switch# show arp inspection statistics vlan
```

```

VLAN Name Forwarded Dropped

```

```

1 DEFAULT_VLAN_1 0 0
200 VLAN200 0 0

```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                                                                                                              |
|---------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show arp state

```
show arp state {all | failed | incomplete | permanent | reachable | stale}
```

## Description

Shows ARP (Address Resolution Protocol) cache entries that are in the specified state.

| Parameter  | Description                                                                                                                                                                                                                                                                                                       |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all        | Shows the ARP cache entries for all VRF (Virtual Router Forwarding) instances.                                                                                                                                                                                                                                    |
| failed     | Shows the ARP cache entries that are in <code>failed</code> state. The neighbor might have been deleted.                                                                                                                                                                                                          |
| incomplete | Shows the ARP cache entries that are in <code>incomplete</code> state. An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. A solicitation request was sent, and the switch is waiting for a solicitation reply or a timeout. |
| permanent  | Shows the ARP cache entries that are in <code>permanent</code> state. ARP entries that are in a permanent state can be removed by administrative action only.                                                                                                                                                     |
| reachable  | Shows the ARP cache entries that are in <code>reachable</code> state, meaning that the neighbor is known to have been reachable recently.                                                                                                                                                                         |
| stale      | Shows ARP cache entries that are in <code>stale</code> state. ARP cache entries are in the <code>stale</code> state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.                                        |

## Examples

```
switch# show arp state failed
```

| IPv4 Address | MAC | Port   | Physical Port | State  |
|--------------|-----|--------|---------------|--------|
| 192.168.1.4  |     | vlan10 |               | failed |

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                                                                                                              |
|---------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show arp summary

```
show arp summary [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows a summary of the IPv4 and IPv6 neighbor entries on the switch for all VRFs or a specific VRF.

| Parameter      | Description                  |
|----------------|------------------------------|
| all-vrfs       | Selects all VRFs.            |
| vrf <VRF-NAME> | Specifies the name of a VRF. |

### Examples

Showing summary ARP information for all VRFs:

```
switch# show arp summary all-vrfs
```

| ARP Entry's State                | : IPv4 | IPv6 |
|----------------------------------|--------|------|
| Number of Reachable ARP entries  | : 2    | 0    |
| Number of Stale ARP entries      | : 0    | 0    |
| Number of Failed ARP entries     | : 2    | 2    |
| Number of Incomplete ARP entries | : 0    | 0    |
| Number of Permanent ARP entries  | : 0    | 0    |

```

Total ARP Entries: 6 : 4 2

```

Showing a summary of all IPv4 and IPv6 neighbor entries on the primary and secondary (peer) switches:

```
vsx-primary# show arp summary
ARP Entry's State IPv4 IPv6

Number of Reachable ARP entries 25858 32231
Number of Stale ARP entries 0 1
Number of Failed ARP entries 0 257
Number of Incomplete ARP entries 0 0
Number of Permanent ARP entries 0 0

Total ARP Entries- 58347 25858 32489

vsx-primary# show arp summary vsx-peer
ARP Entry's State IPv4 IPv6

Number of Reachable ARP entries 25858 32168
Number of Stale ARP entries 0 3
Number of Failed ARP entries 0 317
Number of Incomplete ARP entries 0 0
Number of Permanent ARP entries 0 0

Total ARP Entries- 58346 25858 32488

```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                                                                                                              |
|---------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show arp timeout

```
show arp timeout [<INTERFACE>]
```

## Description

Shows the age-out period for each ARP (Address Resolution Protocol) entry for a port, LAG, or VLAN interface.

| Parameter   | Description                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------|
| <INTERFACE> | Specifies a physical port, VLAN, or LAG on the switch. For physical ports, use the format (for example, 1/3/1). |

## Examples

Showing ARP timeout information for a VLAN:

```
switch# show arp timeout vlan2
Port VRF Timeout

vlan2 default 1800
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                                                                                                              |
|---------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show arp vrf

```
show arp {all-vrfs | vrf <VRF-NAME>}
```

## Description

Shows the ARP table for all VRF instances, or for the named VRF.

| Parameter      | Description                                                           |
|----------------|-----------------------------------------------------------------------|
| all-vrfs       | Specifies all VRFs.                                                   |
| vrf <VRF-NAME> | Specifies the name of a VRF. Length: 1 to 32 alphanumeric characters. |

## Examples

Showing ARP entries for VRF **test**.

```
switch# show arp vrf test
ARP IPv4 Entries:

IPv4 Address MAC Port Physical Port State VRF
10.20.30.40 00:50:56:bd:6a:c5 1/1/29 1/1/29 reachable test

```

```
Total Number Of ARP Entries Listed: 1.
```

```
switch# show arp all-vrfs
ARP IPv4 Entries:
```

| IPv4 Address   | MAC               | Port   | Physical Port | State     | VRF  |
|----------------|-------------------|--------|---------------|-----------|------|
| 192.168.120.10 | 00:50:56:bd:10:be | 1/1/32 | 1/1/32        | reachable | red  |
| 10.20.30.40    | 00:50:56:bd:6a:c5 | 1/1/29 | 1/1/29        | reachable | test |

```
Total Number Of ARP Entries Listed: 2.
```

Showing ARP entries for all VRFs.

```
switch# show arp all-vrfs
ARP IPv4 Entries:
```

| IPv4 Address   | MAC               | Port   | Physical Port | State     | VRF  |
|----------------|-------------------|--------|---------------|-----------|------|
| 192.168.120.10 | 00:50:56:bd:10:be | 1/1/32 | 1/1/32        | reachable | red  |
| 10.20.30.40    | 00:50:56:bd:6a:c5 | 1/1/29 | 1/1/29        | reachable | test |

```
Total Number Of ARP Entries Listed: 2.
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                                                                                                              |
|---------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show ipv6 neighbors

```
show ipv6 neighbors {all-vrfs | vrf <VRF-NAME>}
```

### Description

Shows entries in the ARP table for all IPv6 neighbors for all VRFs or for a specific VRF.

When no parameters are specified, this command shows all ARP entries for the default VRF, and state information for `reachable` and `stale` entries only.

| Parameter                         | Description                                                           |
|-----------------------------------|-----------------------------------------------------------------------|
| <code>all-vrfs</code>             | Specifies all VRFs.                                                   |
| <code>vrf &lt;VRF-NAME&gt;</code> | Specifies the name of a VRF. Length: 1 to 32 alphanumeric characters. |

## Examples

```
switch# show ipv6 neighbors
IPv6 Entries:
```

```

IPv6 Address MAC Port Physical Port State

fe80::a21d:48ff:fe8f:2700 a0:1d:48:8f:27:00 vlan2300 1/1/31 reachable
fe80::f603:43ff:fe80:a600 f4:03:43:80:a6:00 vlan2300 1/1/30 reachable

```

```
Total Number Of IPv6 Neighbors Entries Listed: 2.

```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                          |
|---------------|-----------------------------|------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |

## show ipv6 neighbors state

```
show ipv6 neighbors state {all | failed | incomplete | permanent | reachable | stale}
```

## Description

Shows all IPv6 neighbor ARP (Address Resolution Protocol) cache entries, or those cache entries that are in the specified state.

| Parameter  | Description                                                                                                                                                                                                                                                                                                             |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| all        | Shows all ARP cache entries.                                                                                                                                                                                                                                                                                            |
| failed     | Shows ARP cache entries that are in <code>failed</code> state. The neighbor might have been deleted. Set the neighbor to be unreachable.                                                                                                                                                                                |
| incomplete | Shows ARP cache entries that are in <code>incomplete</code> state. An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. This means that a solicitation request was sent, and you are waiting for a solicitation reply or a timeout. |
| permanent  | Shows ARP cache entries that are in <code>permanent</code> state.                                                                                                                                                                                                                                                       |



| Parameter | Description                                                                                                                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reachable | Shows ARP cache entries that are in <code>reachable</code> state, meaning that the neighbor is known to have been reachable recently.                                                                                                                                         |
| stale     | Shows ARP cache entries that are in <code>stale</code> state.<br>ARP cache entries are in the <code>stale</code> state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly. |

## Example

```
switch# show ipv6 neighbors state all
```

| IPv6 Address              | MAC               | Port   | Physical Port | State     |
|---------------------------|-------------------|--------|---------------|-----------|
| 100::2                    | 48:0f:cf:af:f1:cc | lag1   | lag1          | reachable |
| 300::3                    | 48:0f:cf:af:33:be | vlan3  | 1/4/20        | reachable |
| fe80::4a0f:cfff:feaf:f1cc | 48:0f:cf:af:f1:cc | lag1   | lag1          | reachable |
| 200::3                    | 48:0f:cf:af:33:be | 1/4/11 | 1/4/11        | reachable |
| fe80::4a0f:cfff:feaf:33be | 48:0f:cf:af:33:be | vlan3  | 1/4/20        | reachable |

```
Total Number Of IPv6 Neighbors Entries Listed- 5.
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                                                                                                              |
|---------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

### Accessing Aruba Support

|                         |                                                                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aruba Support Services  | <a href="https://www.arubanetworks.com/support-services/">https://www.arubanetworks.com/support-services/</a>                                                                        |
| Aruba Support Portal    | <a href="https://asp.arubanetworks.com/">https://asp.arubanetworks.com/</a>                                                                                                          |
| North America telephone | 1-800-943-4526 (US & Canada Toll-Free Number)<br>+1-408-754-1200 (Primary - Toll Number)<br>+1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working) |
| International telephone | <a href="https://www.arubanetworks.com/support-services/contact-support/">https://www.arubanetworks.com/support-services/contact-support/</a>                                        |

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

#### Other useful sites

Other websites that can be used to find information:

|                                           |                                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Airheads social forums and Knowledge Base | <a href="https://community.arubanetworks.com/">https://community.arubanetworks.com/</a>                                               |
| Software licensing                        | <a href="https://lms.arubanetworks.com/">https://lms.arubanetworks.com/</a>                                                           |
| End-of-Life information                   | <a href="https://www.arubanetworks.com/support-services/end-of-life/">https://www.arubanetworks.com/support-services/end-of-life/</a> |
| Aruba software and documentation          | <a href="https://asp.arubanetworks.com/downloads">https://asp.arubanetworks.com/downloads</a>                                         |

### Accessing Updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

#### Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

## My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

## Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

## Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

### Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

## Documentation Feedback

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